



When Do Dashboards Override Judgement?

***Decision Scenarios, Data Quality and Managerial Reliance
on Business Intelligence Dashboards***

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Abstract

Most managers in large organisations now begin a decision by looking at a screen. A business intelligence dashboard gathers the numbers a manager is expected to watch, places them on one page and marks some of them red or green. Research shows that managers rely on these screens. Much less is known about the point at which that reliance stops helping judgement and starts replacing it. This study therefore asks in which situations managers put dashboard output ahead of their own judgement, why the pattern persists, and what it costs in decision accuracy and in the range of options considered.

The study takes a critical realist position within a pragmatic orientation and uses a sequential explanatory mixed methods design with three strands. A scenario-based decision experiment with 133 practising managers varied, within participants, whether a data fault was disclosed on the screen and whether the decision was framed as auditable against the system, across eight vignettes each and 1,064 decisions in total. A structured documentary analysis coded twelve organisational failures documented by regulators, statutory inquiries and royal commissions, drawn from nine sectors and four countries. Written interviews with 26 of the experimental participants, selected on how they had behaved in the experiment, were used to explain the mechanism.

Hiding the data fault raised the odds of following the screen against contradicting local evidence (odds ratio 2.69), and framing the decision as auditable raised them again with the data held constant (odds ratio 1.81). The two effects were additive and worked through different channels. Hiding the fault lowered intention to verify and raised confidence, whereas auditable framing changed the choice without moving either. Deference did not reduce the number of alternative actions named. In nine of the twelve documented failures a person questioned the system output and was overruled, and 96 per cent of interviewees agreed that a report-backed decision is easier to defend than a judgement-backed one.

The study separates detection failure, where nobody sees the fault, from authority failure, where somebody sees it but the objection carries no institutional weight, and proposes legitimacy asymmetry as the mechanism that keeps authority failure in place. It concludes with a governance framework of two limbs: detective controls that show provenance and freshness on the face of the screen, and authorising controls that oblige whoever overrules a recorded objection to answer it in writing.

Keywords: business intelligence dashboards; managerial judgement; over-reliance; data quality; accountability; mixed methods.

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Abbreviations

BI	Business intelligence
CAA	Civil Aviation Authority (United Kingdom)
CI	Confidence interval
FCA	Financial Conduct Authority (United Kingdom)
GDPR	General Data Protection Regulation
GEE	Generalised estimating equation
IRR	Incidence rate ratio
KPI	Key performance indicator
MRS	Market Research Society
NATS	National Air Traffic Services (United Kingdom)
OCC	Office of the Comptroller of the Currency (United States)
OR	Odds ratio
OSR	Office for Statistics Regulation (United Kingdom)
PHE	Public Health England
RAG	Red, amber, green (a traffic-light status indicator)
RQ	Research question
SEC	Securities and Exchange Commission (United States)
TSB	TSB Bank plc

Note on Indian units used in the worked example: one lakh is 100,000 and one crore is 10,000,000. Rupee amounts are marked Rs.

Statement on the use of generative artificial intelligence

As permitted by section 4.9 of the ISO8007 guidelines, generative artificial intelligence tools were used to help with document formatting, with drawing figures from coded data, and with checking spelling and grammar. The research design, source selection, coding judgements, analysis and interpretation are my own. Every source cited has been checked against its primary record and is listed with its access route in the reference register described at Appendix K.

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Chapter 1

Introduction

1.1 Background

A manager in a large organisation rarely starts a decision with a blank sheet of paper. He or she starts with a screen. The business intelligence dashboard has moved from being a reporting tool to being the place where decisions begin. It settles which numbers the manager sees, which are hidden inside a total, and which account of the decision will be easiest to defend later.

Business intelligence capability is associated with better decision support and better firm performance, because such systems pull scattered data into one place and put analysis inside the daily routine (Chen, Chiang and Storey, 2012; Elbashir, Collier and Davern, 2008; Seddon et al., 2017; Trieu, 2017). In operations and retail, dashboards are used to watch demand and store performance (Gunasekaran et al., 2017; Kache and Seuring, 2017).

What is less well understood is that a dashboard does not show reality. It composes a version of it. What is displayed, what is aggregated away and what is coloured red decide which knowledge is easy to see and which is not (Pauwels et al., 2009; Sarikaya et al., 2019; Yigitbasioglu and Velcu, 2012). That compression makes a dashboard useful and risky at once, because the screen can take over judgement even when the numbers behind it are incomplete or out of date.

1.2 What a dashboard is, and whom this study is about

Working definition
A dashboard is a single screen that gathers the numbers a manager is expected to watch, updates them on a fixed cycle, and presents most of them as a headline figure with a comparison and a status colour. Common names inside firms are the daily sales screen, the MIS report, the KPI board, the performance portal and the operations wall. The tool, whether Power BI, Tableau or another, is not the point. What matters is that one screen has become the accepted starting point for a recurring decision.

Figure 1.1 labels the parts of a typical screen. Four of them do the behavioural work. Each number is pre-computed, so the manager receives a conclusion rather than evidence (A). A

comparison against last year tells him or her whether to be satisfied (D). A status colour turns that comparison into an instruction (C). The data source, if it is shown at all, appears in small grey type at the foot of the screen.

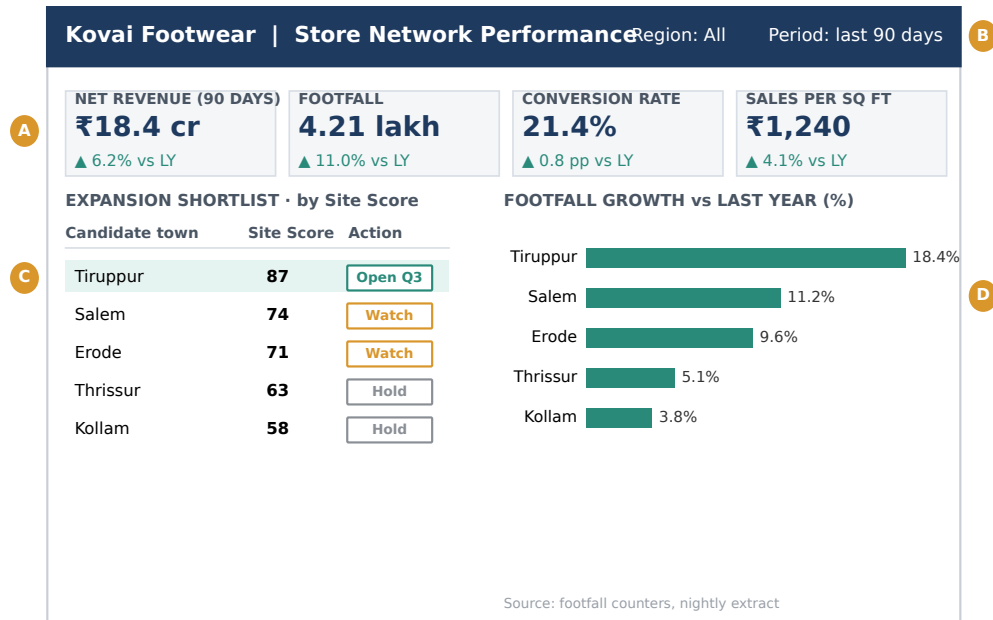


Figure 1.1: The parts of a business intelligence dashboard, drawn from the store network screen used in the worked example at section 1.3. Values are illustrative. A: pre-computed headline figures. B: the scope selectors. C: a ranked shortlist with a status colour and a recommended action. D: a comparison against last year. One lakh is 100,000 and one crore is 10,000,000.

Whom this study is about. The study concerns managers who use a dashboard at least weekly and who make decisions that commit the organisation: opening a site, releasing stock, approving spend, or setting next quarter's plan. It is not about the analysts who build the screen but about the person who must act on it and justify that action later, choosing not only a belief about the numbers but also a story he or she will have to defend (section 2.5).

1.3 A worked example: one store opening decision

One composite case runs through this dissertation as a numbered Thinking Example. It is built from the failure patterns reported at section 4.2. It is used to explain ideas. It is not evidence and it is not counted as such.

THINKING EXAMPLE 1 • Kovai Footwear

The decision, and what the screen did not say

Kovai Footwear runs 22 shoe stores across Tamil Nadu and Kerala from a head office at Coimbatore. Every quarter Ramesh, the regional manager, must recommend where the next store should open, and he starts from the Store Network Performance screen at Figure 1.1.

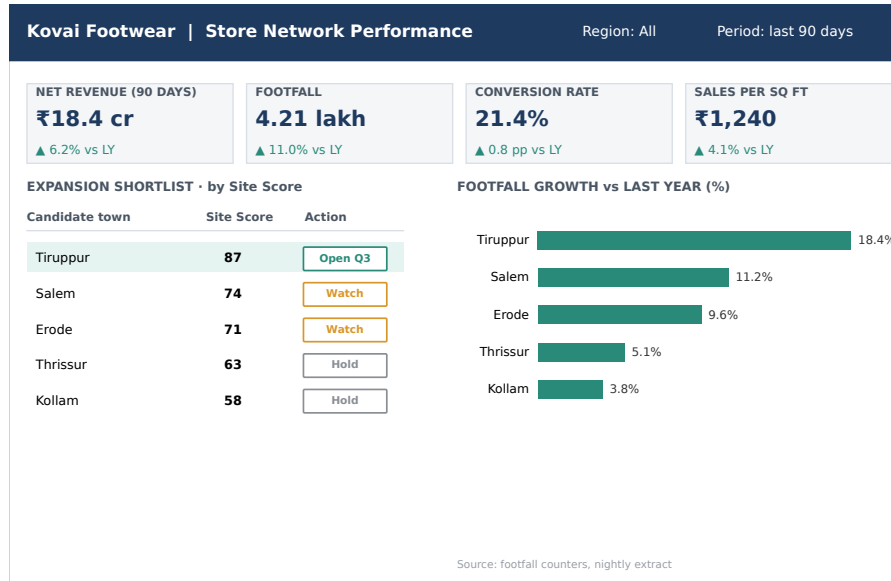
The screen ranks candidate towns by a Site Score that combines footfall growth, average basket value, rent per square foot and competitor density. In June it shows Tiruppur at 87, marked green, with the recommended action “Open Q3”. The next town is thirteen points behind.

Two things were true that the screen did not show. The footfall counter at the Tiruppur site had been offline since 12 May, and the nightly extract carried the last good reading forward, so the number kept looking healthy and no error was displayed. And Meena, the area sales officer who visits the town weekly, had told Ramesh that the crowd near the site is bus-stand traffic rather than shoppers, and that two nearby retailers had cut prices.

Ramesh recommended Tiruppur. The board’s capital expenditure template asks for the Site Score and the payback estimate; it has no field for what Meena observed. The store opened in September, by March was trading at 54 per cent of plan, and the review traced the problem to the stale counter.

Figure 1.2 shows the same screen twice, once with the fault silent and once with a data-provenance banner added. The numbers behind the two screens are identical. What differs is whether the manager has any reason to check, and section 3.5.1 turns that difference into a manipulated experimental condition.

(a) The screen as the regional manager saw it on 3 June



(b) The same screen with a data-provenance banner added

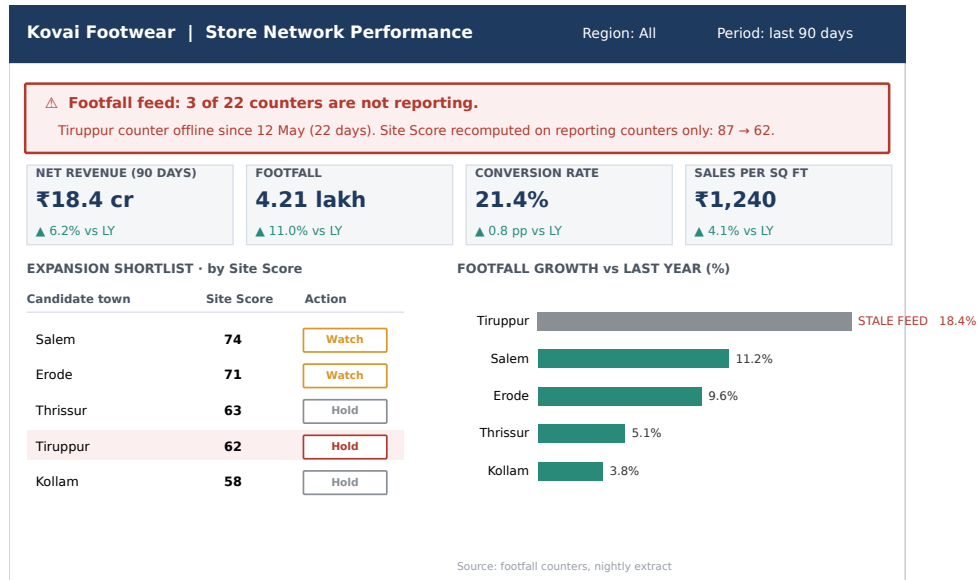


Figure 1.2: The same fault, hidden and disclosed. Panel (a) shows a silent data-quality defect. Panel (b) adds a data-provenance banner, which is the manipulated condition tested by hypothesis H1.

1.4 The research problem

Kovai is a small case with a recoverable loss. The public record shows the same shape without the recovery.

At the Post Office, computer figures were treated as more reliable than the accounts of several hundred sub-postmasters, who were prosecuted on that basis (Post Office Horizon IT Inquiry, 2025). In the Robodebt scheme, averaged income data placed the burden of proof on citizens, and legal advice questioning the lawfulness of the method was set aside

(Royal Commission into the Robodebt Scheme, 2023). At NATS in 2023, one ambiguous flight-planning input propagated into national disruption, and escalation to the engineers who could diagnose it lagged behind the failure (Civil Aviation Authority, 2024).

These are not stories about badly written software. They are stories about a system output becoming more credible inside an organisation than a human objection, which is the shape of the Kovai capital expenditure form.

Research has established that managers rely on analytical outputs, not when that reliance becomes dangerous, or why deference can be rational for the individual even when it is costly for the organisation. The evidence on reliance is mixed (Dietvorst, Simmons and Massey, 2015; Logg, Minson and Moore, 2019), and reviews treat reliance as conditional rather than as a fixed trait (Burton, Stein and Jensen, 2020; Mahmud et al., 2022), so the research question must be conditional as well.

1.5 Research aim and objectives

The aim is to identify when managers put dashboard output ahead of their own judgement, why this persists, and what it costs in decision accuracy and in the options considered. Five objectives follow.

1. To establish experimentally how the visibility of a data fault and the framing of accountability affect the chance of following a dashboard output that conflicts with what the manager knows.
2. To measure whether following the dashboard is associated with naming fewer alternative actions.
3. To characterise, from documented failures, how analytics errors survive undetected and what happened when somebody questioned them.
4. To explain, from managers' own accounts, why dashboard output carries more institutional weight than local knowledge.
5. To develop a governance framework that keeps analytical rigour while protecting the person who disagrees.

1.6 Research questions

Research questions

RQ1 (when, and at what cost). In which decision situations do managers put business intelligence dashboard output ahead of their own judgement, and how does that relate to decision accuracy and to the number of alternatives they generate?

RQ1a How do the visibility of a data fault and the framing of accountability affect the chance of following a dashboard output that conflicts with contextual evidence?

RQ1b Do managers who follow the dashboard generate fewer alternative actions, after allowing for experience and sector?

RQ2 (why). Why do managers do this in those situations, and what organisational conditions keep the pattern in place or interrupt it?

RQ2a What reasons do managers give for following or questioning a dashboard? Is it perceived objectivity, auditability, shortage of time, or the risk of being wrong on their own authority?

RQ2b What conditions of governance, culture or role make disagreement possible?

1.7 Scope and boundaries of the study

Three boundaries apply.

Firstly, the unit of analysis is one decision episode, not the organisation and not the software. Dashboard products are not evaluated, and the return on analytics investment is not measured.

Secondly, the study covers business intelligence dashboards, not generative artificial intelligence, which raises separate questions about fluent but fabricated output. Findings on reliance on such systems are used only where the mechanism is the same (Buçinca, Malaya and Gajos, 2021; Vasconcelos et al., 2023).

Thirdly, the setting is professional decision-making in commercial and public sector organisations. Consumer and clinical settings, where the automation bias literature is better developed (Goddard, Roudsari and Wyatt, 2012; Lyell and Coiera, 2017), are out of scope.

1.8 Significance and contribution

The contribution is a condition-specific account of dashboard reliance. Instead of asking whether analytics adds value, the study asks when analytics creates fragility, and answers from controlled variation, documented failure and the explanations that managers themselves give.

1.9 Structure of the research

Figure 1.3 shows how the remaining chapters fit together.

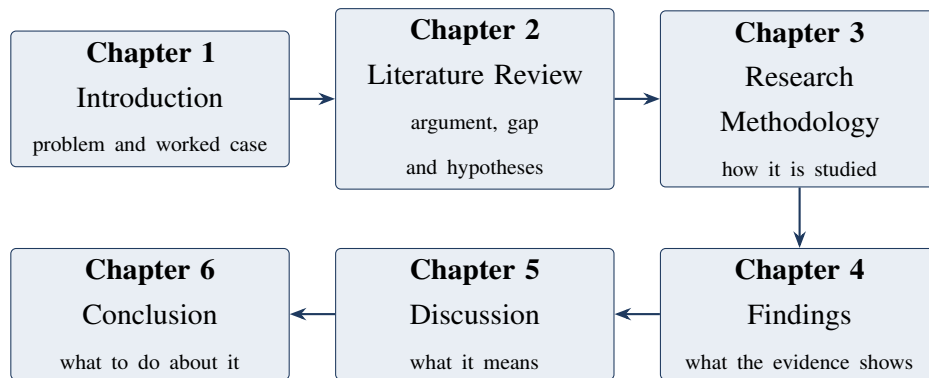


Figure 1.3: Structure of the research

Chapter 2

Literature Review

2.1 Scope and method of the review

The review draws on structured searches of Scopus, Web of Science, Business Source Complete and the ACM Digital Library, with citation tracking forward and backward from the five most central papers. Five search strings covered dashboards and visualisation, reliance on algorithms and automation bias, data quality, creativity and option generation, and the effects of performance management systems. Only peer-reviewed journal articles in English were retained for the main argument, with the window set at 2015 to 2026 and no lower bound for foundational work. Conference proceedings were admitted only for human-computer interaction research, and grey literature only as primary evidence for the documentary strand under the admissibility rule at section 3.5.2. Appendix N documents the searches and Appendix A appraises the ten most central papers.

The review is written as one argument rather than as five separate summaries, because the contribution of this study lies in the links between the literatures. Figure 2.1 sets out the chain in five steps, and section 2.7 asks how far the evidence behind each step can be trusted.

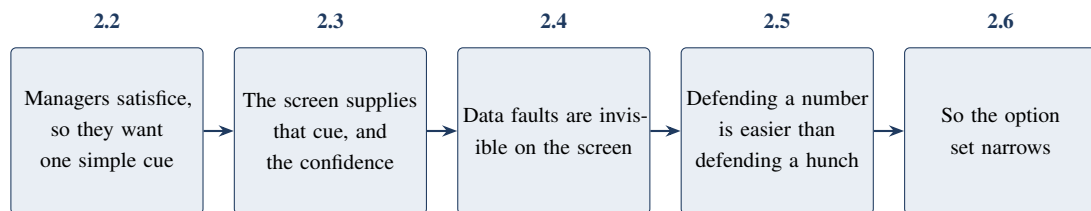


Figure 2.1: The literature review as a single chain of reasoning

2.2 Bounded rationality and the demand for a simple cue

Managers satisfice. Simon (1955) established that decision makers work under limits of information, time and mental capacity, and so adopt strategies that are good enough rather than best. Kahneman (2011) later separated fast automatic thinking from slow effortful thinking, and noted that experience produces reliable intuition only where the feedback from the environment is consistent.

A dashboard is an attractive satisficing device. It turns a problem of many dimensions into a few pre-judged signals at almost no mental cost, and Ramesh in section 1.3 behaved as

Simon would predict. Bounded rationality does not say that managers believe dashboards to be true, only that under time pressure they stop searching once an authoritative-looking cue appears. When that search stops is governed by the design of the dashboard.

2.3 From thinking to interface: layout, prominence and confidence

Layout, metric choice, update frequency and visual encoding all shape how a screen is read (Pauwels et al., 2009; Yigitbasioglu and Velcu, 2012). Sarikaya et al. (2019) found that real dashboards afford different behaviours, from exploration to monitoring, and Bach et al. (2023) catalogued the design patterns that direct attention. Graphical perception research supplies the mechanism: the encoding determines what a reader notices and how accurately it is read (Cleveland and McGill, 1984; Tufte, 2001; Ware, 2013). Simon explains why a manager accepts a simplified cue, and visualisation research explains which cue it will be, namely the most prominent one, and how confident he or she will feel about it.

Automation is misused when its output is accepted too readily (Parasuraman and Riley, 1997), even against contradictory evidence (Mosier et al., 1998; Skitka, Mosier and Burdick, 1999). Reliance reduces how often the raw evidence is sampled (Parasuraman and Manzey, 2010), and the harder a system is to verify, the more readily it is trusted (Goddard, Roudsari and Wyatt, 2012; Lyell and Coiera, 2017).

Current work rejects any simple story of over-trust. An algorithm that is seen to err loses its users sharply (Dietvorst, Simmons and Massey, 2015), whereas other tasks show a preference for algorithmic over human judgement (Logg, Minson and Moore, 2019). Reviews conclude that reliance varies with the task, the perceived expertise of the system, the visibility of its errors and the control the person keeps (Burton, Stein and Jensen, 2020; Mahmud et al., 2022), which is why RQ1 is framed conditionally. Over-reliance falls when a cognitive forcing function raises the cost of accepting an output without thinking (Buçinca, Malaya and Gajos, 2021), and explanations reduce it only when reading them is easier than doing the task (Vasconcelos et al., 2023). Reliance is therefore a design and governance variable rather than a personality trait.

2.4 Why data quality is a behavioural problem

Data quality is defined along dimensions such as accuracy, timeliness, completeness, consistency and relevance (Wang and Strong, 1996), and it must be judged in context, because information that exists may still be unfit for a particular decision (Pipino, Lee and Wang, 2002; Strong, Lee and Wang, 1997). Poor data quality is expensive (Redman, 1998), it is a managerial rather than a technical responsibility, and most organisations do not measure it (Nagle, Redman and Sammon, 2020). Decision quality depends on the

whole chain from source to decision (Janssen, van der Voort and Wahyudi, 2017), and problems upstream compound invisibly downstream (Sambasivan et al., 2021).

What that literature misses is that data faults are dangerous because they are invisible on the screen. An out-of-date metric, a partial extract or a changed definition renders as cleanly as a correct figure.

The core prediction of this study follows. Harm from poor data quality is proportional not to the size of the error but to the confidence with which the error is displayed, which the signal visibility manipulation at section 3.5.1 tests. Where a fault is disclosed, the cost of checking falls and questioning should rise. Where it is hidden, the manager has no cue to trigger the slower thinking that Kahneman describes.

2.5 Legitimacy asymmetry

Individual thinking alone cannot explain why deference persists, because experienced managers often do suspect that a screen is wrong. The further explanation is institutional.

Decision support systems are not neutral. They take part in how an organisation knows things and they reshape its decision routines (Arnott and Pervan, 2005; Shollo and Galliers, 2016). Performance measurement systems produce well-documented side effects, including gaming and the replacement of judgement by whatever happens to be measurable (Franco-Santos and Otley, 2018). Espeland and Sauder (2007) showed reactivity at work: once a measure carries public consequences, organisations rearrange themselves around it, and it begins to create the reality it claims to describe.

From these strands the study takes its central idea, which it calls legitimacy asymmetry. A decision backed by a dashboard is standardised and easy to defend after the event. An objection based on local knowledge is specific to its situation and easy to dismiss. The asymmetry is not about which account is more accurate. It is about which account is more defensible.

Algorithmic systems act as instruments of organisational control, not only as sources of information (Kellogg, Valentine and Christin, 2020). Professionals switch off their own judgement where questioning an opaque system is professionally risky (Lebovitz, Lifshitz-Assaf and Levina, 2022). The design problem is how much authority should sit with the system and how much with the person (Shrestha, Ben-Menahem and von Krogh, 2019).

Legitimacy asymmetry explains what individual cognition cannot: deference can be rational for the manager even when it is costly for the organisation. It also yields a testable prediction, hypothesis H2. If a decision is framed as auditable against the system, deference should rise even though the data have not changed.

2.6 Option-set narrowing and creative exploration

Creativity here means how many workable alternatives a manager generates under uncertainty. Evaluation pressure and rigid performance systems reduce creative output (Amabile, 1998). Organisational creativity depends on the person, the group and the setting (Woodman, Sawyer and Griffin, 1993), combines originality with usefulness (Runco and Jaeger, 2012) and is contingent on context (Anderson, Potočnik and Zhou, 2014).

The mechanism proposed here is option-set narrowing. A dashboard does not forbid alternatives. It makes some of them visible and already costed while leaving others unmeasured and harder to argue for. Under time pressure the unmeasured option is not rejected; it is never written down.

Two recent findings make this an empirical question. Artificial intelligence capability is positively associated with organisational creativity, which suggests that analytics can widen exploration (Mikalef and Gupta, 2021). Generative artificial intelligence raises individual creative output but reduces the diversity of ideas across people (Doshi and Hauser, 2024). The direction is conditional, so this study measures the alternatives generated alongside deference rather than assuming either.

THINKING EXAMPLE 2 • Kovai Footwear

The five steps, applied to one Monday morning

2.2 Simple cue. Ramesh has an hour before the capital expenditure call. The Site Score of 87 arrives pre-judged, and his search stops there.

2.3 Salience and confidence. The 87 is the largest number in the top row, with a green light beside it. The green does not summarise the position; it reports that somebody has already finished thinking about it.

2.4 Silent fault. The counter is dead, yet the screen prints 18.4 per cent growth in the same typeface, colour and position as a live figure.

2.5 Legitimacy asymmetry. The form has a field for the Site Score and none for what Meena saw. If the store fails after Ramesh follows the score, the system was wrong. If it fails after he overrules it on Meena's word, he was wrong.

2.6 Option-set narrowing. A kiosk in Erode at a fifth of the capital cost, or a delay of two quarters, would both be sensible. Neither is on the screen, so neither reaches the board paper.

2.7 Critical appraisal of the evidence base

Four weaknesses in this evidence base shaped the design that follows.

The reliance experiments are not run on managers. The foundational studies use student samples on generic forecasting tasks (Dietvorst, Simmons and Massey, 2015; Logg, Minson and Moore, 2019), which removes the variable this study treats as decisive, accountability. Such findings show that reliance can be shifted, not how large it is at work.

This study therefore recruits practising managers and manipulates accountability.

The automation bias evidence comes from a narrow set of domains. The strongest evidence is from aviation and clinical decision support (Goddard, Roudsari and Wyatt, 2012; Lyell and Coiera, 2017; Mosier et al., 1998), which have formal escalation and logging. Managerial use has neither, so managers may question more freely because no procedure constrains them, or less freely because none protects them.

The data quality literature measures dimensions, not behaviour. The dimensional frameworks (Pipino, Lee and Wang, 2002; Strong, Lee and Wang, 1997; Wang and Strong, 1996) were built for assessment programmes, not to predict what a decision maker does when a dimension degrades. Recent managerial work reports how little firms measure rather than how people respond (Nagle, Redman and Sammon, 2020). Hypothesis H1 addresses this gap.

The analytics value studies cannot show causation. Work linking analytics capability to firm performance rests mainly on cross-sectional surveys in which both are self-reported (Gunasekaran et al., 2017; Mikalef and Gupta, 2021; Wamba et al., 2017), so it is open to common method bias and to reverse causation, since firms already doing well can afford better analytics (Podsakoff et al., 2003).

The four weaknesses point one way. The literature is strong on mechanism and weak on organisational condition.

2.8 Synthesis, research gap and conceptual framework

Each link in Figure 2.1 has been established separately. What has not been done is to examine them as one chain, under controlled conditions, with working managers. Dashboard design research studies comprehension rather than error under conflicting evidence, and creativity is rarely measured in the same study as accuracy and reliance.

The gap addressed by this study

No integrated empirical account states the conditions under which reliance on a business intelligence dashboard turns from a decision aid into a decision risk, or explains why organisations keep that shift in place once it has happened.

Figure 2.2 sets out the initial conceptual framework. Four conditions act on attention and confidence, which in turn shape three outcomes. Two of the conditions, the data quality signal and the accountability regime, are manipulated in the experiment. Dashboard design is held constant across scenarios, and the decision context is measured as perceived stakes rather than manipulated. Figure 2.3 places the study among the five literatures that feed it. Section 5.2 revises the framework in the light of the evidence.

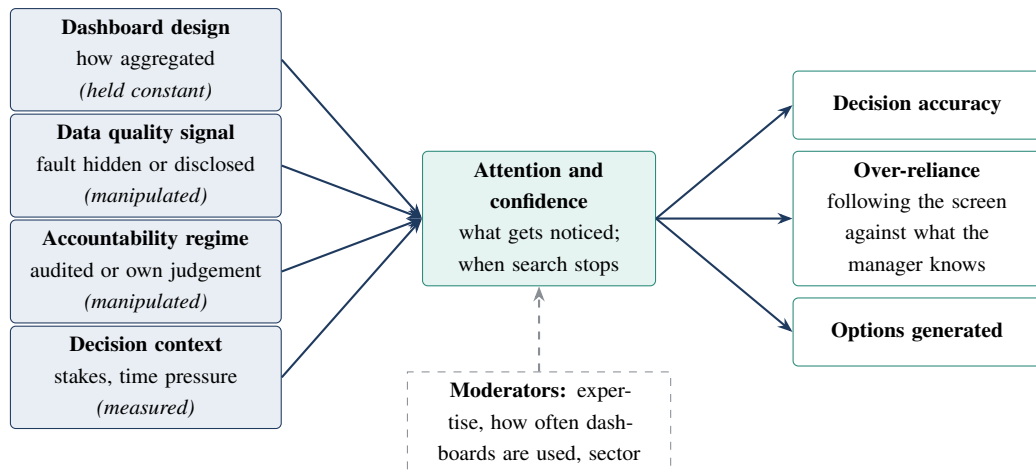


Figure 2.2: Initial conceptual framework. Four conditions act on attention and confidence, which shape three outcomes. RQ1 tests the paths; RQ2 explains why the accountability path behaves as it does.

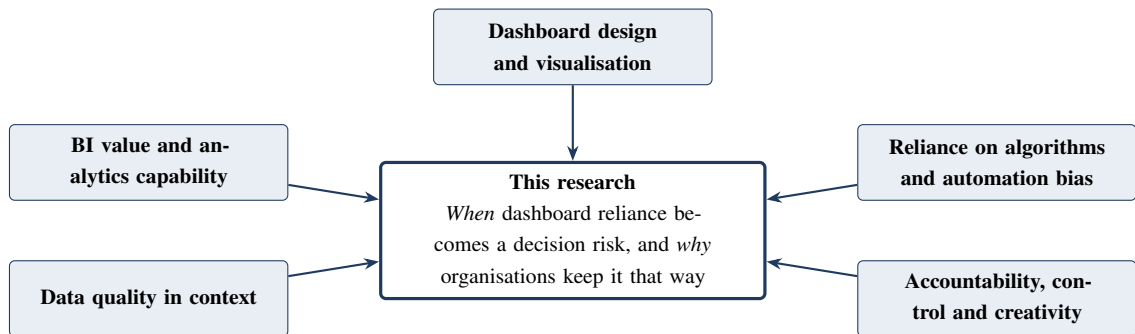


Figure 2.3: The five literatures that feed this study, and where it sits between them

2.9 Hypotheses

The experiment tests the hypotheses in Table 2.1. H3 is stated as two competing predictions, because the literature supports both directions. H5 is declared exploratory in advance because it is a between-participants moderator of a within-participant effect, and the power estimates at Appendix E show that even the within-participant interaction reaches a power of only about .40. The status of each hypothesis was fixed before analysis, because a hypothesis quietly demoted after the data arrive is not evidence of anything.

Table 2.1: Hypotheses, theoretical basis and status in the design

ID	Hypothesis	Theoretical basis	Status
H1	Deference is higher when the data fault is hidden than when a freshness warning is shown.	Wang and Strong (1996); Lyell and Coiera (2017)	Tested within participants
H2	Deference is higher when the decision is framed as auditable against the system than as the manager's own judgement, with the data held constant.	Espeland and Sauder (2007); Franco-Santos and Otley (2018)	Tested within participants
H3a	Higher perceived stakes increase deference, because the manager wants a defensible position.	Legitimacy asymmetry, section 2.5	Descriptive. Stakes measured, not manipulated
H3b	Higher perceived stakes decrease deference, because the manager checks more carefully.	Dietvorst, Simmons and Massey (2015)	As H3a
H4	Managers who defer generate fewer distinct alternative actions.	Amabile (1998); Doshi and Hauser (2024)	Tested; see section 4.3.8
H5	Expertise moderates H1, so that more experienced managers question a hidden-fault dashboard more often.	Kahneman (2011); Lebovitz, Lifshitz-Assaf and Levina (2022)	Exploratory. Underpowered by design; see Appendix E

Chapter 3

Research Methodology

This chapter sets out how the research questions at section 1.6 are answered, following the research onion of Saunders, Lewis and Thornhill (2019) at Figure 3.1, with the choice made in this study written beneath each layer.

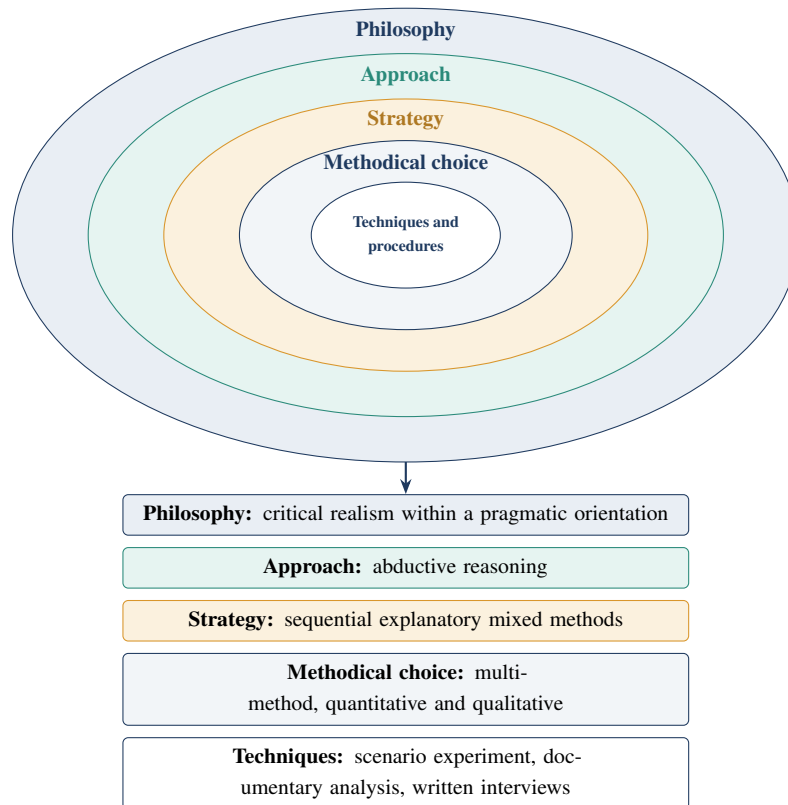


Figure 3.1: The research onion of Saunders, Lewis and Thornhill (2019), with the choice made in this study shown beneath each layer

3.1 Research philosophy: critical realism within a pragmatic orientation

The study takes a critical realist ontology inside a pragmatic orientation. Decision errors and their consequences are real. The mechanisms that produce them, such as legitimacy and professional risk, are socially made and can be reached only by interpretation. That stratified view (Bhaskar, 2008) suits a study whose central idea is a mechanism rather than a variable.

Positivism captures the experiment but not legitimacy, and interpretivism on its own rules

out the controlled comparison that RQ1 needs. Pragmatism is the practical orientation, following standard mixed methods guidance (Creswell and Plano Clark, 2018; Tashakkori and Teddlie, 2010). The full comparison of the positions considered is at Appendix B.

3.2 Research approach: abductive

The study reasons abductively. Theory sets the initial conditions and the hypotheses, and the observed patterns then revise the framework (Saunders, Lewis and Thornhill, 2019). Section 5.2 shows this in operation.

3.3 Research strategy: sequential explanatory mixed methods

3.3.1 Design alternatives considered and rejected

Five alternatives were considered and rejected, with reasons at Appendix D. Figure 3.2 shows the adopted design. Strand 2, the documentary analysis, ran first because it needs neither ethical approval nor recruitment, and its results shaped the interview protocol.

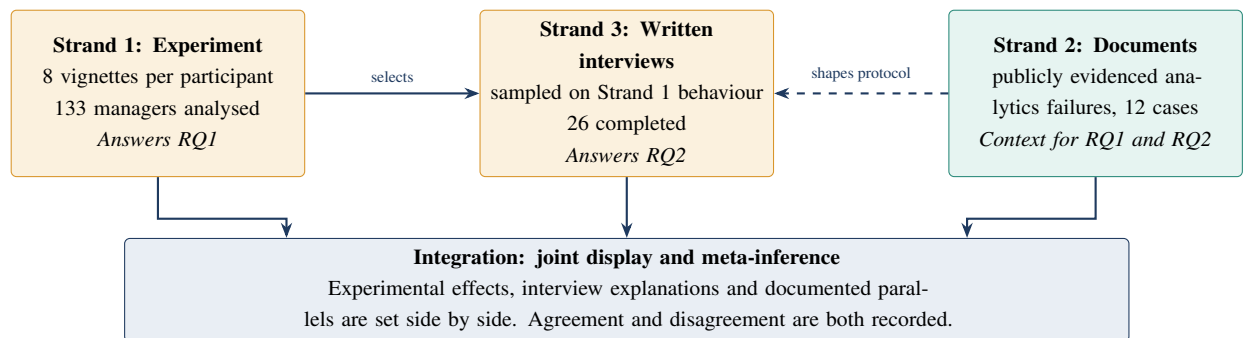


Figure 3.2: Sequential explanatory design with a concurrent documentary strand

3.3.2 Mapping of research questions to methods

Table 3.1 was fixed before data collection. It maps each research question onto the method, the data source and the analysis used.

Table 3.1: Mapping of research questions to method, data source, analysis and expected output

RQ	Method	Data source	Analysis	Expected output
RQ1a	Scenario experiment (Strand 1)	Managers, 8 vignettes each, target of at least 640 decision observations	GEE logistic regression clustered on participant; odds ratios with 95% CI	Chance of deference by condition
RQ1b	Same instrument, count outcome	Number of distinct alternative actions per scenario	Poisson or negative binomial regression	Estimate of option-set narrowing
RQ1 (con-text)	Documentary analysis (Strand 2)	Regulator notices, inquiry and royal commission reports	Structured extraction; cross-case matrix	Failure pattern matrix
RQ2a	Written interviews (Strand 3)	Managers selected on Strand 1 behaviour, floor of 8	Reflexive thematic analysis	Reasons for deferring and for questioning
RQ2b	Interviews and documents	Interview accounts set against control failures	Pattern matching across strands	Conditions that permit challenge; governance framework

3.4 Research methods: mixed method

The experiment gives numerical data on behaviour under varied conditions, which suits RQ1. The documents and the interviews explain the mechanism behind that behaviour, which suits RQ2. Neither part is subordinate to the other, and integration takes place at section 4.5.

3.5 Data collection

3.5.1 Strand 1: scenario-based decision experiment

Instrument design. Experimental vignette methodology suits judgement research where the factors of interest cannot be manipulated in the field (Aguinis and Bradley, 2014; Atzmüller and Steiner, 2010). Each scenario shows a realistic dashboard together with a short brief that carries one piece of local evidence conflicting with the headline signal; the Kovai scenario at section 1.3 is one of the eight. Following section 2.7, the participants are practising managers rather than students, and accountability framing is manipulated rather than held constant. The factors, levels and outcome measures are at Appendix E and the instrument at Appendix O.

The experimental design

A 2×2 within-subjects design, data quality signal (fault hidden or warning shown) by accountability framing (auditable against the system or the manager's own judgement), with each cell seen twice, so eight vignettes per participant. Scenario j takes condition $(j + V) \bmod 4$, where V is drawn at random from 0 to 3, so that across the four versions of the instrument every scenario appears in every cell exactly once. Scenario is therefore orthogonal to condition and can enter the model. Presentation order is shuffled independently. All eight scenarios are written at comparable stakes, and perceived stakes is measured after each vignette rather than manipulated.

Figure 3.3 shows the path a participant takes through the instrument.

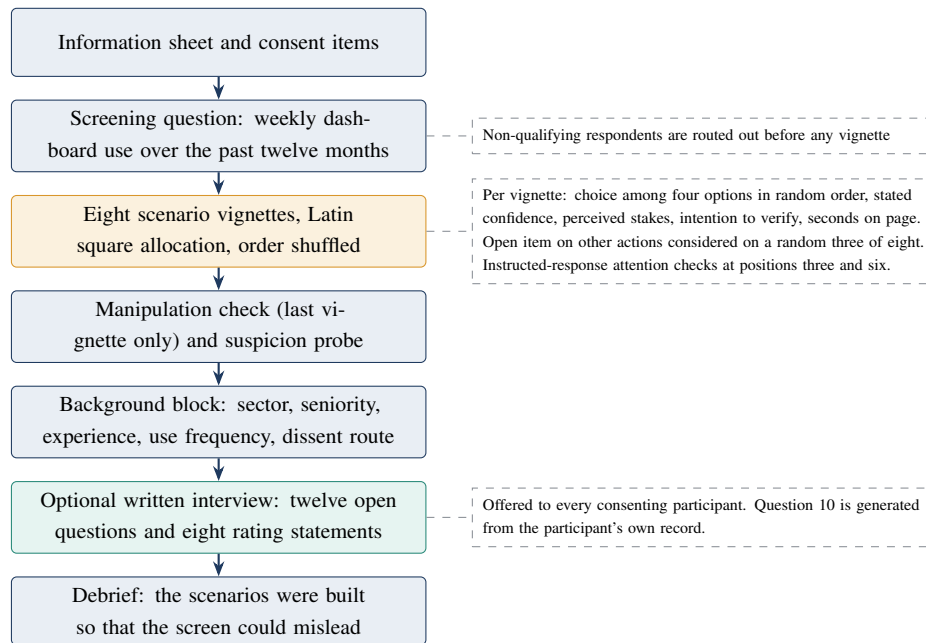


Figure 3.3: The participant's path through the online instrument

Sample size, from Monte Carlo estimation rather than a rule of thumb. The conventional two-proportion calculation (Cohen, 1988; Field, 2018) does not apply here, because no closed form exists for a clustered binary outcome with a crossed scenario effect. Power was therefore estimated by Monte Carlo replication, 400 replications per cell, of the model actually used, and the grid is at Appendix E. Power tracked the total number of observations almost exactly and was close to indifferent to how they were split between participants and vignettes: 480 observations returned a power of about .74 whether from 120 participants seeing four vignettes, 80 seeing six or 60 seeing eight. The target was set at 80 analysable participants, giving 640 observations and a power of about .87 for the main effects.

Sauer et al. (2011) found response consistency degrading beyond ten vignettes per respondent, so eight sits below the lowest count they tested; section 4.3.2 reports the order

check. The power grid also carries a warning that is reported rather than suppressed: the interaction reaches a power of only about .40 even at 120 participants seeing eight vignettes each, and a between-participants moderator such as H5 is weaker still. Both are declared exploratory in advance.

The sample. Participants are working professionals who use dashboards or business intelligence tools in a managerial or analytical role, recruited purposively through professional networks with snowball extension, on free routes only. A screening question enforces weekly dashboard use over the past twelve months, and non-qualifying respondents are routed out before any vignette is shown.

Measures and safeguards. Each vignette records the choice among four options presented in random order, stated confidence, perceived stakes, intention to verify and seconds on page. The open item on other actions considered is asked on a random three vignettes of the eight. Two instructed-response attention checks sit at positions three and six. A manipulation check follows the last vignette only, so that it cannot cue earlier trials, and a closing suspicion probe identifies participants who inferred the design.

Screening rule, fixed before analysis. A participant is excluded for failing either attention check, for a median page time below fifteen seconds, or for a total instrument time below five minutes.

Analysis. The primary model is a generalised estimating equation with a logit link, an exchangeable working correlation clustered on participant and robust standard errors, predicting deference from the data quality signal, the accountability framing and their interaction. A population-averaged model is preferred to a random-intercept model because the quantities of interest are marginal effects across managers. Option counts are modelled by Poisson regression with overdispersion tested. The pre-committed fallback, if the clustering assumptions fail, is a paired within-participant comparison by exact binomial test. Pre-specified robustness runs cover an independence working correlation, exclusion of participants who guessed the design, and a covariate-adjusted model with scenario fixed effects.

3.5.2 Strand 2: documentary analysis

Rationale and admissibility. Documentary analysis suits a phenomenon that leaves a durable and verifiable trace where the researcher cannot be present (Bowen, 2009; Krippendorff, 2018). Only documents whose independent existence can be verified are admissible: regulator enforcement notices and consent orders, statutory inquiry and royal commission reports, regulator-commissioned reviews, audit office and parliamentary reports, and listed company disclosures. Press coverage may locate a case but is never the basis for a coded judgement. Cases were chosen for maximum variation across sector and

failure type, avoiding an all-finance sample, subject to the event falling between 2010 and 2026, documentation in English, and settled rather than contested facts.

Evidential grading. Grade A is a case documented by a dedicated statutory investigation, royal commission or enforcement notice; Grade B is official organisational disclosure with parliamentary or regulatory scrutiny but no dedicated independent investigation.

Coding. Each case was coded against a fixed frame of eight dimensions, reproduced at Appendix F, with a verbatim quotation and page reference recorded for every judgement. The coding was carried out by the researcher alone. The page-referenced quotations make every judgement checkable against its source, and section 6.5 records the absence of an inter-coder agreement statistic as a limitation.

3.5.3 Strand 3: written interviews

Format. The interviews are asynchronous written interviews: twelve open questions delivered in the same online instrument, immediately after the experiment or later by invitation, with eight rating statements and a background block. They were offered to every consenting participant.

The written format has a cost and a benefit. It costs probing, so a thin answer cannot be followed up, and it costs tone and hesitation. It buys access to busy managers, who complete the form when they choose, and the participant's own words rather than a transcription, which removes one layer of researcher mediation. The format suits a study of what people will commit to writing about their own organisation, not a study of affect. The instrument is at Appendix L.

Behavioural sampling. Interview participants are drawn from Strand 1 respondents who consent to follow-up, so the sampling is behavioural rather than convenience based. Question 10 is generated for each participant from their own record. It selects one vignette, preferring one where the fault was hidden and the participant followed the screen, quotes the participant's own choice back to them, and asks what would drive that decision at their own workplace, so the link between behaviour and account is auditable rather than asserted.

Number. A floor of eight was set, since saturation in a homogeneous sample usually appears within about twelve interviews (Guest, Bunce and Johnson, 2006), a number feasible in the time available (Bryman, 2016).

Analysis. Coding follows reflexive thematic analysis (Braun and Clarke, 2006). Codes came from a first reading of the answers to the questions on doubt, escalation and consequence, and each code carries an explicit decision rule in the coding script, so that counts are reproducible and can be reapplied by a second coder. Section 4.4.5 reports the join to

experimental behaviour.

THINKING EXAMPLE 3 • Kovai Footwear

Which strand would have caught which part

The case is one episode, so no single method could recover all of it. That is the reason for using three.

Strand 1, the experiment, would catch the effect of the dead counter. Running the same decision with and without the banner of Figure 1.2(b) isolates what the missing warning does on its own. It says nothing about the capital expenditure form.

Strand 2, the documents, would never contain Kovai at all. A store at 54 per cent of plan produces no inquiry and no enforcement notice, which is the survivorship problem of section 5.4 in one line.

Strand 3, the interviews, is the only strand that reaches the form. Asking Ramesh what happened after he passed Meena's report upward is the only way to learn that there was nowhere for it to go.

3.6 Quality criteria

Quantitative and qualitative criteria are treated separately, following the usual convention (Lincoln and Guba, 1985; Saunders, Lewis and Thornhill, 2019). For Strand 1, internal validity rests on randomised and counterbalanced allocation within participants, construct validity on pilot testing and the manipulation check, and protection against common method bias on the outcome being a behavioural choice while the predictors are manipulated (Podsakoff et al., 2003). For Strand 2, authenticity rests on the admissibility rule and the evidential grading. For Strand 3, credibility rests on behaviour-linked sampling and member checking of the summarised themes. On reflexivity, my analytics training biases me towards reading deference as error, so the coding deliberately looks for cases where deference turned out to be right. The full mapping is at Appendix C.

3.7 Ethical considerations

Ethical approval, the fieldwork risk assessment and the GDPR data management assessment were approved before any primary data collection began, under the University Code of Good Practice in Research. The study follows the MRS Code of Conduct (Market Research Society, 2019) on informed consent, voluntary participation and the avoidance of harm. The completed forms, the participant information sheet and the consent form are reproduced at Appendix Q.

Participants must not feel personally judged. The scenarios have better and worse answers, and participants are professionals whose competence is part of their identity. The study is therefore presented as research into how dashboards are designed and used, never

into individual ability. No participant receives a score, and results are reported only in aggregate. Some concealment is unavoidable if the behaviour is to be observed at all. It is handled by disclosure at debrief, which explains that the scenarios were built so that the dashboard could mislead and that following it was reasonable, rather than by active deception.

Commercial confidentiality. Interviewees work in commercially sensitive environments, so the information sheet bars the disclosure of confidential information, proprietary dashboard designs and named clients. Any such passage is excluded from the transcript, and employers are described by sector and size band only.

Data protection. No name, email address or gender is collected in the experiment. Participants who opt into follow-up provide a contact address, which makes them pseudonymous rather than anonymous, so contact details are held in a separate table that can be deleted on its own. Data are held on University encrypted storage, interview text is pseudonymised at export, and processing is by consent under the UK GDPR. Participants may withdraw without giving a reason and may request deletion up to two weeks after taking part, after which pseudonymisation prevents retrieval. The information sheet at Appendix Q describes an online conversation; the interview strand was delivered instead in written form inside the same instrument, with the consent items, rights of withdrawal and confidentiality bar carried across unchanged, so no recording was made.

The dignity of those harmed. Strand 2 uses only public material, and several cases involve identifiable people who suffered serious harm, most obviously in the Post Office and Robodebt cases. The analysis therefore concentrates on system-level control failures rather than on individual conduct, and treats the human consequences with care.

3.8 Methodological limitations

Four limitations apply. Firstly, the scenario design measures stated decisions in a low-stakes setting. No participant loses money or standing by choosing wrongly, so the accountability manipulation imitates accountability rather than imposing it, and the bias runs towards understating deference. Secondly, purposive recruitment limits generalisation and may over-represent managers who already think about analytics. Thirdly, the documentary strand suffers from severe survivorship bias, discussed at section 5.3. Fourthly, the strands do not carry equal evidential weight: authoritative documents but few of them, a larger experimental sample but an artificial task, and rich interview accounts that rely on recall, so integration must be interpretive rather than arithmetic.

Chapter 4

Findings

4.1 Overview

Section 4.2 reports the documentary strand, section 4.3 the experiment and section 4.4 the written interviews. Section 4.5 brings the three together.

4.2 Strand 2: documentary analysis of twelve organisational failures

4.2.1 The case sample

Twelve cases from nine sectors and four countries met the inclusion criteria and were coded in full. Appendix I lists each case with its period, evidential grade and primary source under the admissibility rule at section 3.5.2.

4.2.2 Failure types and whether the fault could be seen

Figure 4.1 shows the failure types split by whether the fault could be seen at the point of use.

Model-logic failures dominate, at eight of the twelve cases. In each of these the system did what it was built to do, on a rule that did not fit the situation. Averaging annual income across fortnights in the Robodebt scheme was arithmetically correct and substantively wrong for anyone whose earnings varied (Royal Commission into the Robodebt Scheme, 2023).

More important for this study, the fault was silent in seven cases and only partly visible in four more. It was plainly visible to those affected in one case only, the TSB migration of 2018 (Financial Conduct Authority, 2022), which supports the premise at section 2.4 that data and model faults do not announce themselves.

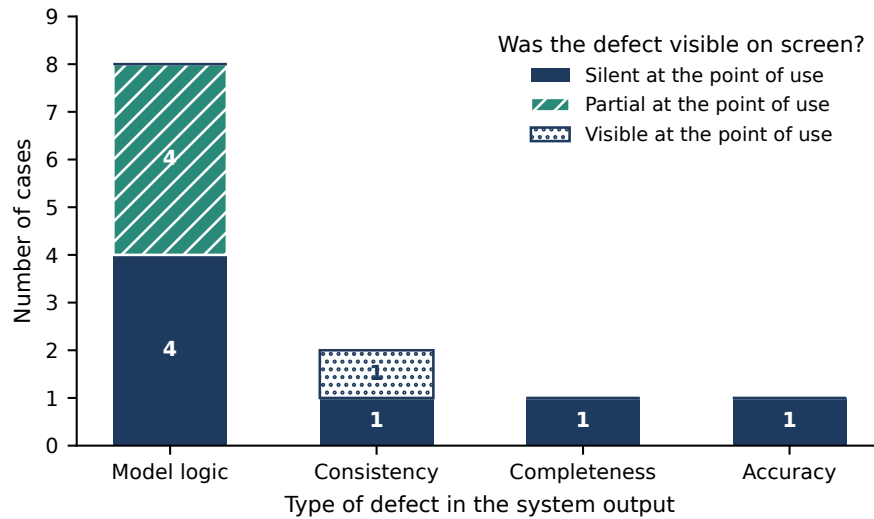


Figure 4.1: Types of fault across the twelve cases, split by whether the fault could be seen on the screen at the point of use

4.2.3 How long the fault survived

Figures 4.2 and 4.3 plot the time taken to notice each fault against whether there was a signal.

Silent faults survived for years, about sixteen in the case of Post Office Horizon (Post Office Horizon IT Inquiry, 2025). Where the fault produced a visible sign, correction was much faster: forty-five minutes for Knight Capital (Securities and Exchange Commission, 2013) and a few hours for the NATS failure (Civil Aviation Authority, 2024). The pattern is not clean: the Public Health England undercount was coded silent yet was caught within about a week, and Zillow was coded partly visible yet ran for about six months, so visibility shortens the lag without determining it. Exposure time is also confounded with visibility, since silent faults tend to occur in slow administrative processes and signalled faults in real-time systems. No significance test is offered on twelve purposively chosen cases; the record is consistent with visibility governing the speed of correction, which H1 tests under controlled conditions.

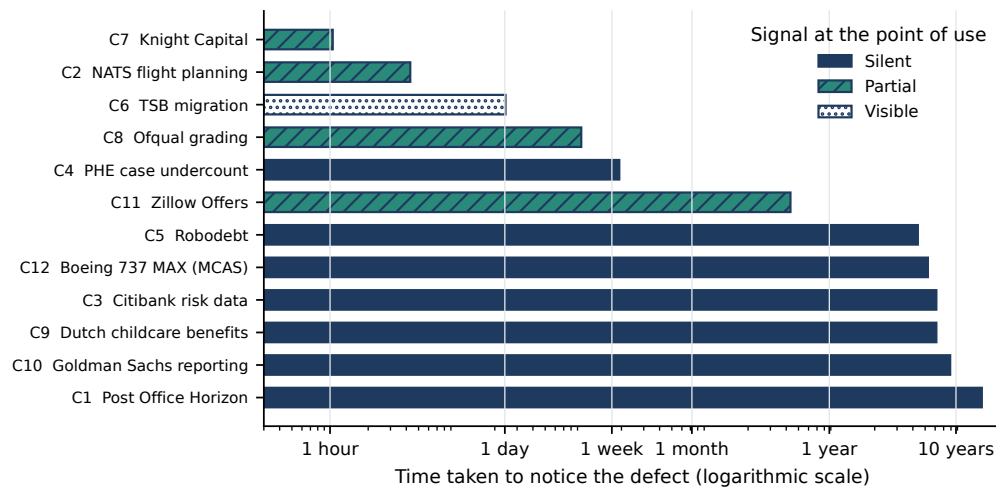


Figure 4.2: Time taken to notice each fault, ordered shortest to longest and shaded by whether the fault was visible at the point of use

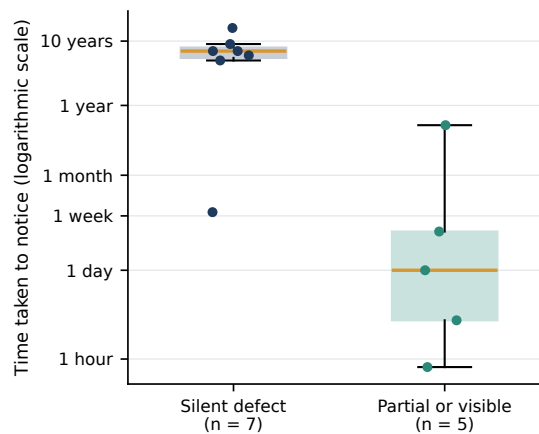


Figure 4.3: Time taken to notice, for silent faults against partly or fully visible ones, with individual cases shown

4.2.4 The main finding: somebody did object, and was overruled

In nine of the twelve cases a person questioned the system output, and the objection was overruled, ignored or left unactioned. An objection was acted on in time in one case only, when Zillow closed its iBuying operation (Zillow Group, 2021). In two cases nobody objected, and in both of those the fault was silent.

Figure 4.4 cross-tabulates the outcome of the challenge against the coded strength of legitimacy asymmetry. Every case with a strong asymmetry falls in the raised-and-overruled group.

Sub-postmasters questioned the Horizon figures for years and were prosecuted rather than investigated (Post Office Horizon IT Inquiry, 2025). Engineering concerns about the 737 MAX depending on a single sensor did not change the certification outcome (US House

Committee on Transportation and Infrastructure, 2020). Knight Capital differs in kind: its systems generated ninety-seven automated emails inside the relevant window, but these were not treated as alerts (Securities and Exchange Commission, 2013).

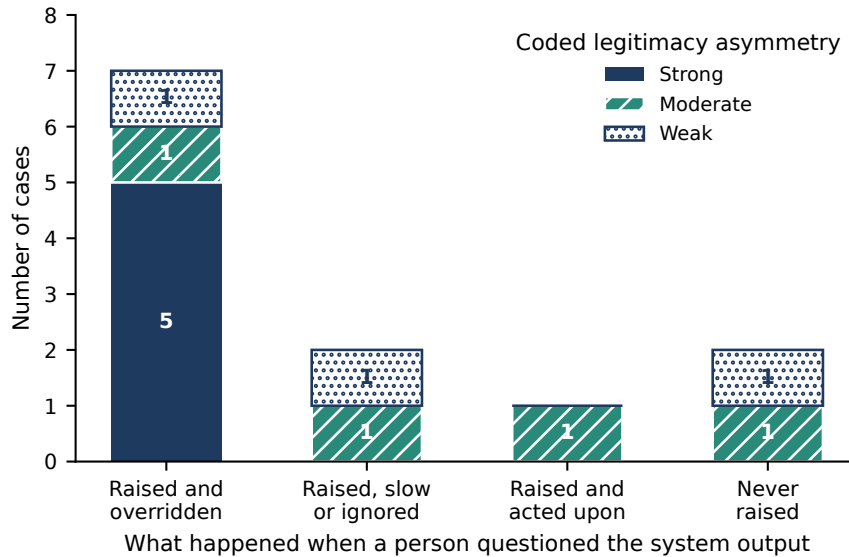


Figure 4.4: What happened when somebody questioned the system output, cross-tabulated with the coded strength of legitimacy asymmetry

4.2.5 Cross-case matrix

Figure 4.5 shows the coded matrix for the four dimensions that make up the mechanism at section 2.5, and Appendix J gives six of the eight coded dimensions in full. Horizon, Robodebt, the Dutch childcare benefits scheme and the 737 MAX each combine a silent fault, a strong legitimacy asymmetry and a suppressed objection. They differ only on the control: in Horizon one was present but ineffective, whereas in the other three it was absent or bypassed.

C1 Post Office Horizon	High	High	High	Med
C2 NATS flight planning	Med	Med	Med	Med
C3 Citibank risk data	High	Med	High	Med
C4 PHE case undercount	High	Med	Low	High
C5 Robodebt	High	High	High	High
C6 TSB migration	Low	Low	High	Med
C7 Knight Capital	Med	Low	Med	High
C8 Ofqual grading	Med	High	High	Med
C9 Dutch childcare benefits	High	High	High	High
C10 Goldman Sachs reporting	High	Low	Low	Med
C11 Zillow Offers	Med	Med	Low	Low
C12 Boeing 737 MAX (MCAS)	High	High	High	High
	Defect was silent	Legitimacy asymmetry	Challenge was suppressed	Control absent or bypassed

Figure 4.5: The four main coded dimensions across the twelve cases. Horizon, Robodebt, the Dutch childcare scheme and the 737 MAX agree on the first three dimensions and differ on the fourth.

4.2.6 Controls, and why they failed

A control was present and effective in one case only. In the Public Health England undercount nobody reconciled the records sent against the records received, so a silent truncation at the row limit of an obsolete file format went unnoticed for about a week. Every control that failed was a detective control, built for spotting problems, rather than an authorising control, built for acting on a person who had spotted one. Section 6.2 has to accommodate that observation.

THINKING EXAMPLE 4 • Kovai Footwear

The twelve cases, read back onto one store

Kovai is not one of the twelve. A store at 54 per cent of plan produces no royal commission. But the signature at Table J.1 is the same at both scales.

Silent fault. Seven of the twelve cases carried no visible sign and survived for years. The Tiruppur counter carried no visible sign and had been dead for 22 days when Ramesh made his recommendation.

Objection raised and overruled. In nine of the twelve cases somebody said something and it changed nothing. Meena said something and it changed nothing. The difference is scale, not shape, and at Kovai as in the twelve the only control that existed was one for spotting problems, not one for acting on a person who had spotted one.

4.3 Strand 1: experimental findings

4.3.1 Sample and screening

The export holds 147 participant records and 1,176 observations, eight per participant, with no missing vignettes. The screening rule at section 3.5.1 removes 14 participants (9.5 per cent), all for failing an instructed-response check and none by the speed filters alone. The analytic sample is therefore 133 participants and 1,064 observations, above the target of 80. Figure 4.6 shows the flow. Median time on a scenario page is 75 seconds (interquartile range 54 to 102), and median time on the whole instrument is 17.1 minutes, longer than the information sheet stated; section 6.5 returns to that point.

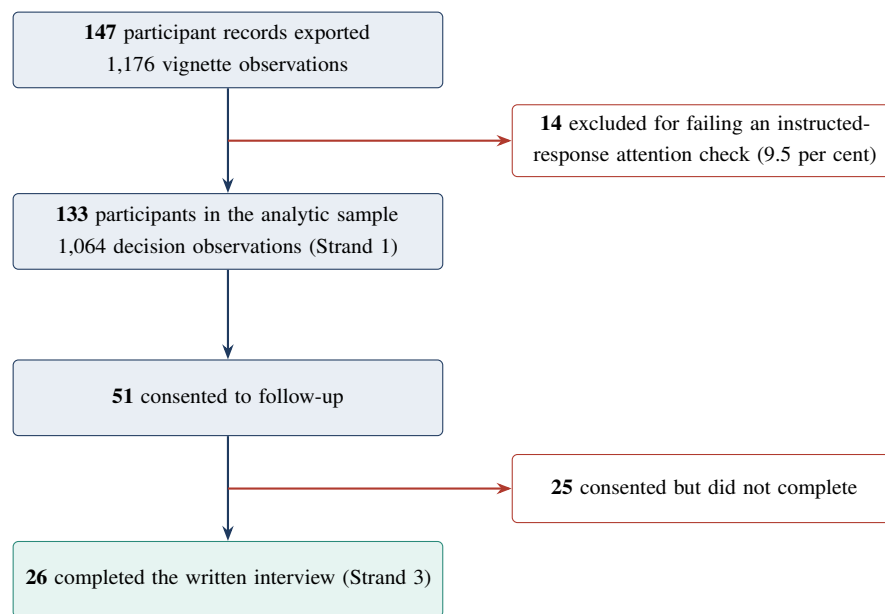


Figure 4.6: Participant flow from the exported records to the analytic sample and the interview strand

Table 4.1 profiles the analytic sample, which is not claimed to be representative of managers in general. Only 29 of the 133 (21.8 per cent) say that their organisation has a formal route for recording disagreement with a report that is actually used.

Table 4.1: Profile of the analytic sample ($N = 133$)

Sector	<i>n</i>	%	Seniority and experience	<i>n</i>	%
Technology and software	23	17.3	<i>Seniority</i>		
Manufacturing and industrial	23	17.3	Manager	41	30.8
Banking and financial services	22	16.5	Team lead or assistant manager	25	18.8
Retail and consumer goods	13	9.8	Senior manager or head of function	25	18.8
Consulting and professional services	11	8.3	Analyst or executive	24	18.0
Logistics, transport and supply chain	9	6.8	Director or above	18	13.5
Education	8	6.0	<i>Experience</i>		
Energy and utilities	8	6.0	Less than 2 years	10	7.5
Healthcare and life sciences	7	5.3	2 to 4 years	23	17.3
Public sector and government	6	4.5	5 to 9 years	39	29.3
Other	3	2.3	10 to 14 years	30	22.6
			15 years or more	31	23.3

4.3.2 Checks on the design

Four checks preceded the hypothesis tests. **Balance:** every participant contributed two observations in each of the four cells, and across the four versions every scenario appeared in every cell, so condition is orthogonal to scenario. **Order:** deference does not drift across the eight positions (OR 1.01 per position, 95% CI [0.95, 1.07], $p = .727$), although median page time falls from 95 seconds to 65, so eight vignettes sit within this population's tolerance. **Manipulation check:** 78.9 per cent correctly identified whether the last vignette carried a notice, but only 29.3 per cent gave the correct retrospective count of four, which is consistent with shallow rather than absent processing. **Suspicion:** 40.6 per cent described a pattern resembling the design, and excluding them strengthens the effects (section 4.3.9).

4.3.3 Descriptive results

Across the 1,064 observations, participants followed the dashboard against the local evidence in 0.280 of decisions and chose the defensible action in 0.408. Figure 4.7 shows the four conditions.

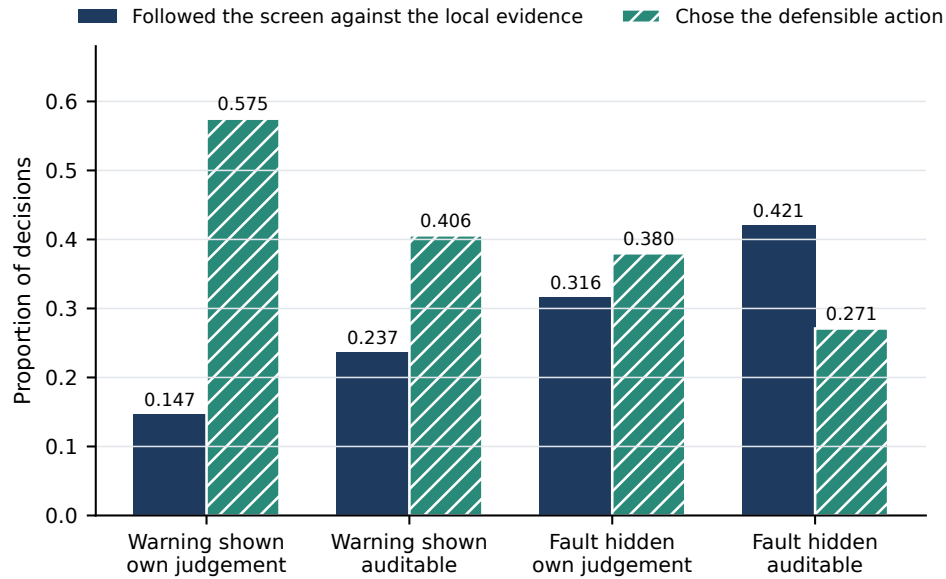


Figure 4.7: Deference and defensible choice across the four conditions

Deference nearly doubles when the fault is hidden rather than disclosed (0.368 against 0.192), and it rises by about half when the decision is framed as auditable rather than as the manager's own (0.329 against 0.231). Defensible choice moves the opposite way on both factors, from 0.491 to 0.325 and from 0.477 to 0.338.

The mean conceals a divided sample: of the 133 participants, 17 never deferred and 7 deferred six times or more, and the mean participant deferred 2.24 times out of eight.

4.3.4 Hypothesis tests

The primary model is the generalised estimating equation pre-committed at section 3.5.1, with a logit link, an exchangeable working correlation clustered on participant and robust standard errors. The within-participant correlation is 0.129, close to the .18 assumed in the power estimation at Appendix E.

H1 is supported. Hiding the fault raises the odds of deference (OR 2.69, 95% CI [1.77, 4.08], $p < .001$). **H2 is supported.** Auditable framing raises the odds with the data held constant (OR 1.81, 95% CI [1.21, 2.70], $p = .004$). **The two effects are additive, not interactive.** The interaction is not distinguishable from zero (OR 0.87, 95% CI [0.50, 1.51], $p = .628$). Section 2.7 treated these as inputs to a single attentional pathway; section 5.2 takes up their independence.

The result survives the pre-committed fallback: within participants, 76 deferred more often under the hidden condition against 20 the other way, with 37 tied (exact binomial, $p < .001$). Figure 4.8 collects the estimates and Appendix P gives the full models.

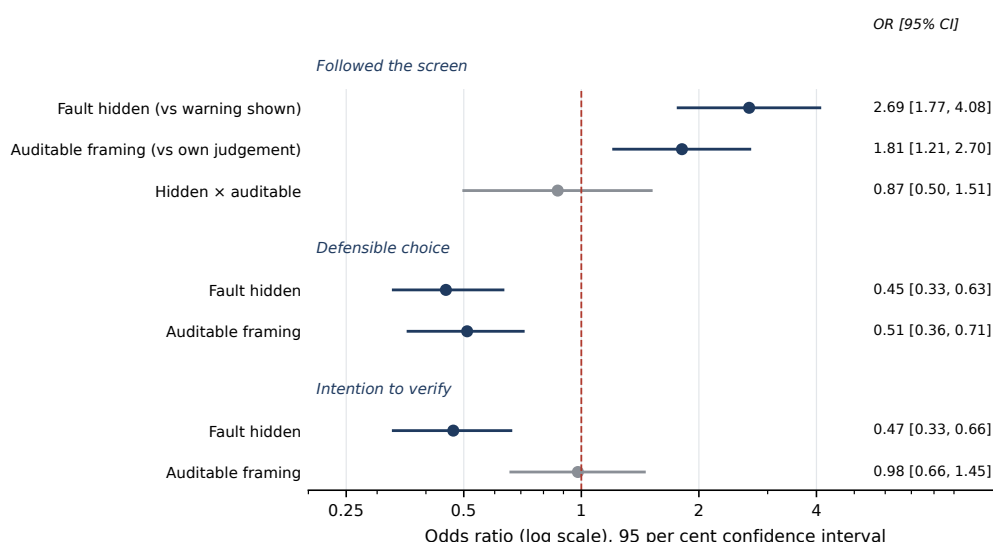


Figure 4.8: Odds ratios with 95 per cent confidence intervals for the three outcomes

4.3.5 Separating detection failure from authority failure

Section 4.2.4 showed from the documents that two distinct things go wrong where the framework at section 2.8 had treated them as one. Figure 4.9 measures the two separately by decomposing deference from the baseline condition.

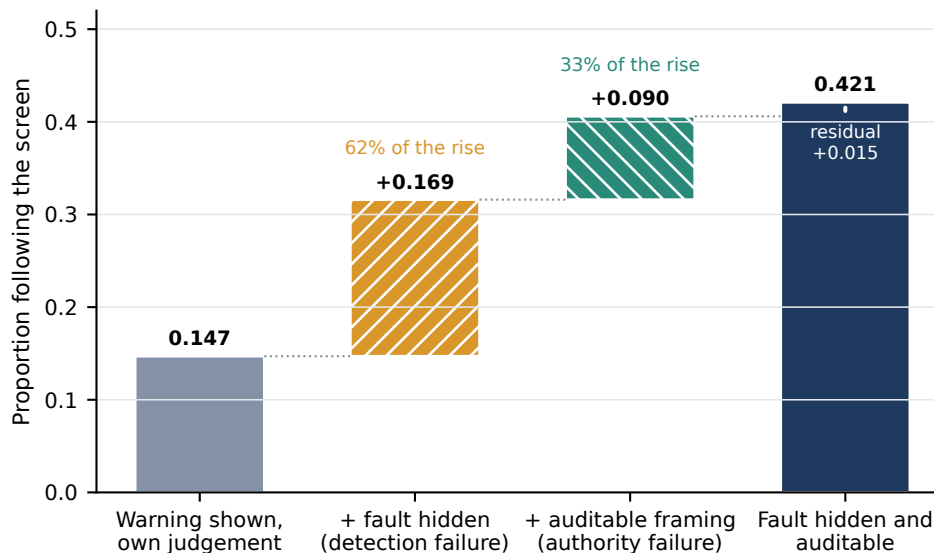


Figure 4.9: Decomposition of deference from the disclosed, own-judgement baseline

Where the warning is shown and the decision is the manager's own, 0.147 of decisions follow the screen. Hiding the fault adds 0.169 and auditable framing adds 0.090. Together they give 0.421, against 0.406 under independence, so the residual is 0.015 and the additive account holds. Detection failure accounts for about 62 per cent of the rise above baseline and authority failure for about 33 per cent.

Neither mechanism depends on the other. Auditable framing raises deference whether the warning is shown (OR 1.81) or the fault is hidden (OR 1.58, 95% CI [1.13, 2.19], $p = .007$), and hiding the fault raises it under own judgement (OR 2.69) and under audit (OR 2.34, 95% CI [1.68, 3.27], $p < .001$).

4.3.6 The two manipulations work through different channels

Additivity shows that two effects exist. It does not show that they differ in kind. The instrument measures intention to verify and stated confidence alongside the choice, and Figure 4.10 shows what each manipulation moves.

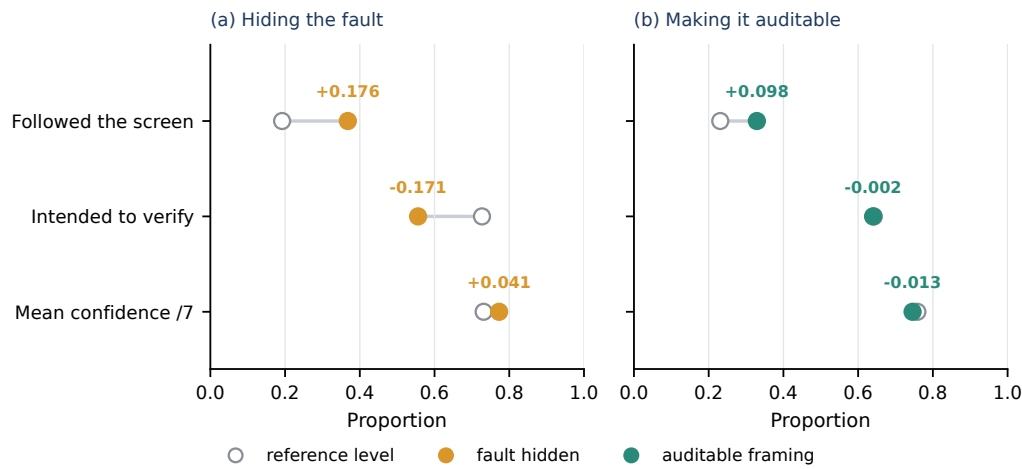


Figure 4.10: What each manipulation moves. Hiding the fault shifts all three measures; auditable framing shifts only the choice.

Hiding the fault behaves as an attentional account predicts. Deference rises, intention to verify falls from 0.727 to 0.556 (OR 0.47, 95% CI [0.33, 0.66], $p < .001$), and stated confidence rises from 5.12 to 5.41 on a seven-point scale. The absence of a warning is read as an assurance, which restates the point made by Wang and Strong (1996) about believability, now as behaviour rather than as a rating.

The accountability frame behaves differently. Deference rises by 0.090, but intention to verify does not move (OR 0.98, 95% CI [0.66, 1.45], $p = .924$) and confidence does not move either.

This is the discriminating result: an attentional explanation predicts that people under audit would look less carefully, and they did not.

4.3.7 Accuracy, confidence and calibration

Both manipulations reduce the chance of the defensible choice: hiding the fault (OR 0.45, 95% CI [0.33, 0.63], $p < .001$) and auditable framing (OR 0.51, 95% CI [0.36, 0.71], $p < .001$). Confidence carries some signal, at 5.42 when the defensible action was chosen

against 5.15 when it was not, and the difference holds when condition is controlled (+0.31, $p < .001$). Figure 4.11 shows the pattern.

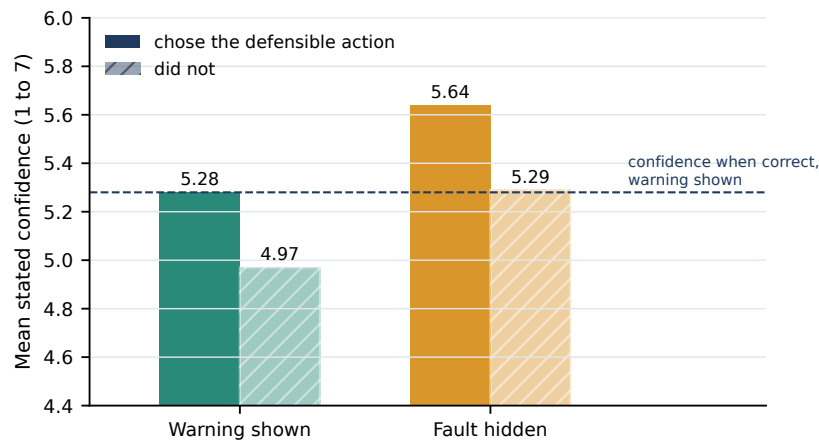


Figure 4.11: Stated confidence by condition and by whether the defensible action was chosen

Under a hidden fault, participants who chose wrongly reported a confidence of 5.29, as high as participants who chose correctly when the warning was shown (5.28). Hiding the fault raises confidence by +0.34 points ($p < .001$) whether or not the decision was right.

4.3.8 Option generation: H4 is not supported

H4 predicted that managers who defer would name fewer alternative actions. They do not. The open item was asked on three vignettes of eight, giving 399 responses from all 133 participants, with a mean of 1.88 distinct actions. A Poisson model clustered on participant returns an incidence rate ratio for deference of 0.94 (95% CI [0.80, 1.10], $p = .413$). The dispersion statistic is 0.95, so no negative binomial fallback was needed. Neither manipulation moves the count (Figure 4.12).

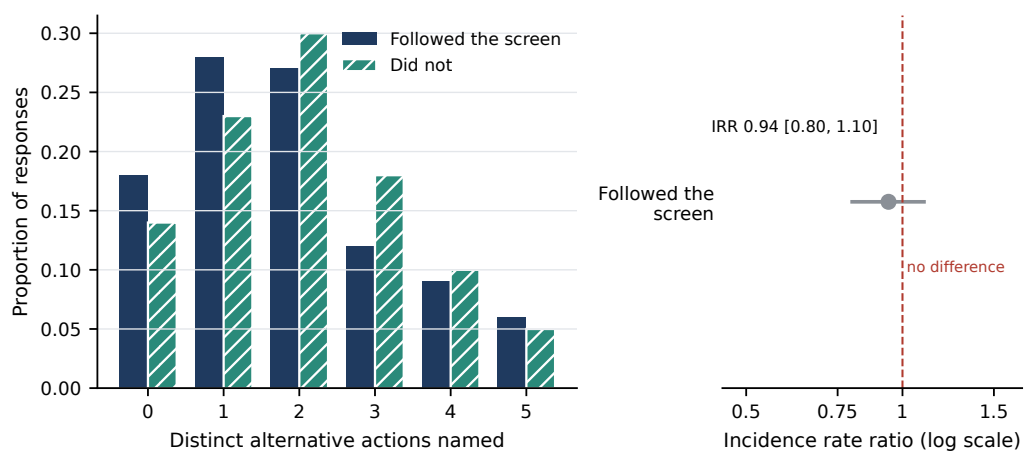


Figure 4.12: Distinct alternative actions named, with the model estimate at right

Three readings survive the data: the free-text box may be an insensitive proxy for the

option set a manager works through over a week; the effect may be real but small, since the interval does not exclude a reduction of a fifth; or narrowing may operate at the level of the team rather than the individual, which section 4.4 suggests and a scenario task cannot reach. The claim is retained only where the interview evidence carries it.

4.3.9 Expertise, stakes and robustness

H5, declared exploratory. Experience neither moderates the hidden-fault effect (OR 1.03, 95% CI [0.83, 1.28], $p = .782$) nor has a main effect of its own (OR 0.86, 95% CI [0.69, 1.08], $p = .203$). Appendix E puts the power of even the within-participant interaction at about .40, and a between-participants moderator is weaker still, so this null is close to uninformative and is not evidence of no moderation.

Perceived stakes. Higher perceived stakes is associated with slightly less deference (OR 0.83, 95% CI [0.71, 0.98], $p = .029$), which points to H3b rather than H3a. The range is narrow, with a mean of 5.91 and a standard deviation of 0.87 and only 65 observations at four or below, so a manipulation would be needed to settle it.

The formal-route item predicts nothing. Participants who say that a formal route exists and is used defer at 0.267 against 0.284 for everyone else ($p = .755$), and the item adds nothing to the model (OR 0.94, 95% CI [0.60, 1.47], $p = .782$). A tick-box captures whether a policy document exists, not whether an objection is answered; section 4.4.5 asks the second question.

Robustness. Estimates are unchanged under an independence working correlation. Excluding the 54 participants who described the design leaves 79 and strengthens both effects (hidden OR 3.68, auditable OR 2.17), so neither is an artefact of guessing what was being tested. Adjusting for perceived stakes, experience, seniority, use frequency and scenario fixed effects leaves the estimates essentially unchanged (hidden OR 2.85, auditable OR 1.99). Time on page does not predict deference, so the effect is not rushing.

4.4 Strand 3: interview findings

4.4.1 Participants and material

Of the 133 participants, 51 consented to follow-up and 26 completed the written interview. That is above the floor of eight set at section 3.5.3 and inside the range at which saturation usually appears in a homogeneous sample (Guest, Bunce and Johnson, 2006). The median transcript runs to 956 words across the twelve open questions. Ten sectors are represented, none with more than five participants, and seniority runs from analyst to director. Because Question 10 quotes each participant's own experimental choice back to them, the join at section 4.4.5 is possible.

4.4.2 The blame gap, measured directly

Table 4.2 gives the eight rating statements, and Figure 4.13 shows the percentage agreeing with each.

Table 4.2: Rating statements, means and percentage agreeing (6 or 7 on a seven-point scale), $n = 26$

ID	Statement	Mean	SD	% agree
R1	Go with the dashboard when it conflicts with ground knowledge	3.81	1.36	31
R2	Can tell from the screen whether the data is current	3.36	0.91	12
R3	A report-backed decision is easier to defend than a judgement-backed one	5.81	0.75	96
R4	Follow the dashboard, bad outcome: personally held responsible	3.23	0.95	12
R5	Overrule the dashboard, bad outcome: personally held responsible	5.46	0.76	92
R6	Disagreement with a report gets recorded somewhere	3.58	1.50	31
R7	Someone must formally answer a raised concern before the decision proceeds	3.04	1.62	19
R8	Working from dashboards narrows the options the team considers	4.85	0.88	69

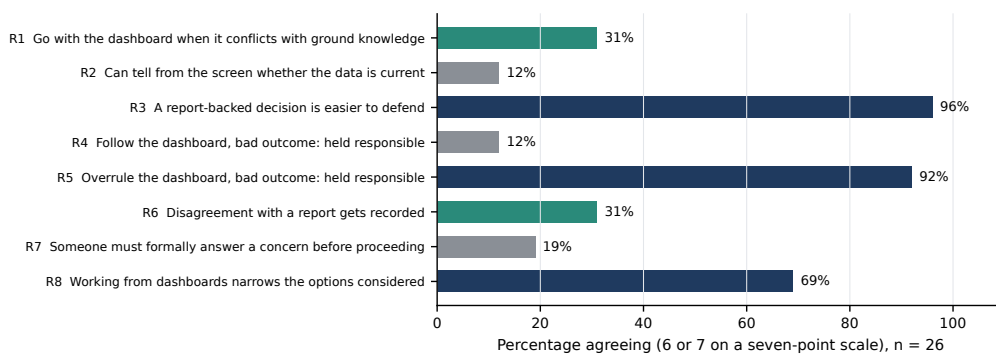


Figure 4.13: Percentage of interview participants agreeing with each rating statement

Three statements carry the argument. 96 per cent agree that a report-backed decision is easier to defend than a judgement-backed one, which is the legitimacy asymmetry of section 2.5 in the participants' own words. R4 and R5 put a number on it: 92 per cent expect to be held personally responsible for overruling the dashboard and being wrong, against 12 per cent for following it and being wrong. The mean gap is 2.23 points (paired $t(25) = 7.56$, $p < .001$, $d = 1.48$), positive for 81 per cent of participants. R7 is the lowest-rated statement, with 19 per cent agreeing, and the proposal at section 6.2 is built to change that.

4.4.3 Themes

Four themes follow, with a fifth recorded but not developed: thirteen of the 26 transcripts describe tools that display nothing when a feed has failed, which is the detection side of the argument in the participants' own words.

The asymmetry is described, not merely implied (21 of 26). Participants distinguish the two ways of being wrong without being prompted, as a system problem against a personal one. One public sector participant explained that a decision taken as per the monitoring data is accepted at every level, whereas a deviation invites the question of what basis the officer had, which is hard to answer even when the officer was right.

The objection is raised and leaves no trace (18 of 26). The technical fault is usually fixed while the report carrying the wrong number is not, and 8 participants describe that split. One retail participant noted that nothing records that a store's conversion metric was invalid for a month, so next year's analysis will read the wrong figure as real.

Recording is not answering (4 of 26). One participant described a properly minuted objection with a file number, traceable years later, which was answered by the statement that the system was working as designed and then closed. Their summary, that recording is not answering, is the sentence this study carries into section 6.2.

Defensive documentation (8 of 26). Managers describe writing a timestamped note before deviating, or securing a superior's agreement by email, to convert a personal decision into a shared one. One said that this is not dishonest, it is how you survive. A cheaper move appears in 6 transcripts: the concern is voiced aloud and never written, which preserves the ability to say it was raised while leaving nothing on record.

4.4.4 Disconfirming accounts

Nine of the 26 transcripts describe the asymmetry as absent or reversed, and they identify when the mechanism switches off. They come from six sectors and share a structure: deviating cautiously is the defensible act, and the reason is institutional rather than cultural.

A healthcare participant signs the staff rota personally, so following the staffing model and having a patient harmed is their responsibility, whereas an extra nurse invites at most a question about agency cost. A banking participant described a committee where following the model without challenge is itself an audit finding, and noted that this holds because it is a regulatory expectation, not because the institution is virtuous.

The rating scale corroborates the coding: the reversal group records a blame gap of 0.78 points against 3.00 for the rest ($t = 5.03, p < .001$).

One account cuts against the proposal at section 6.2. A technology director described an objection that was minuted and answered, but attributed the outcome to their own

seniority, observing that the formal route works well for the people who least need it. Seven transcripts carry that code.

4.4.5 Linking what managers say to how they behaved

Because the interview participants also completed the experiment, each theme can be set against the same participant's own deference rate (Figure 4.14).

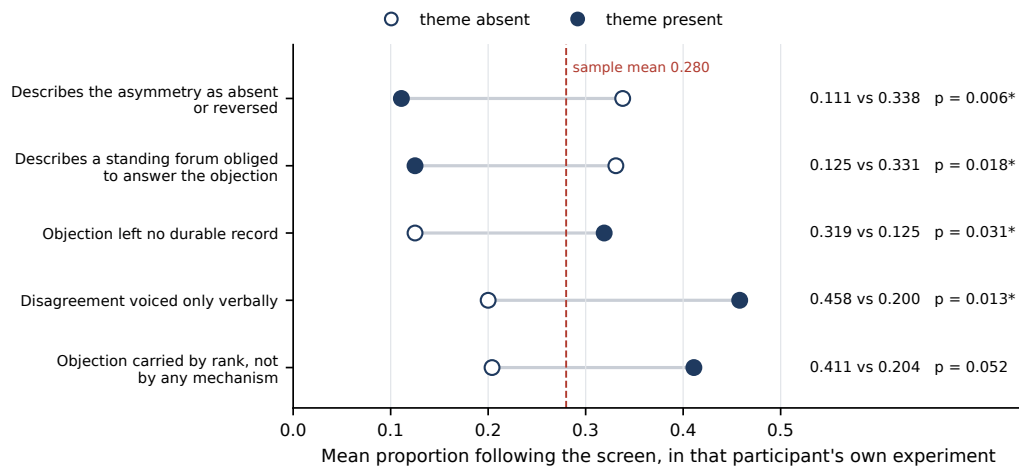


Figure 4.14: Interview themes against the same participant's deference rate. Mann-Whitney U on participant-level rates, $n = 26$.

Participants who describe the asymmetry as absent or reversed deferred at 0.111 against 0.338 for the rest ($p = .006$). Those who describe a standing forum obliged to answer an objection deferred at 0.125 against 0.331 ($p = .018$). In the other direction, participants whose objection left no durable record deferred at 0.319 against 0.125 ($p = .031$), and the six who describe disagreement voiced only verbally deferred at 0.458, the highest of any group ($p = .013$). The ratings agree. The blame gap correlates with the deference rate at $\rho = 0.62$ ($p < .001$), R7 at $\rho = -0.66$ ($p < .001$) and R6 at $\rho = -0.69$ ($p < .001$).

Unlike the tick-box item at section 4.3.9, which predicted nothing, these measures ask whether an objection gets *answered*, not whether a route *exists*.

4.5 Integration across the three strands

Table 4.3 is the joint display. It records disagreement as well as agreement. Table 4.4 then summarises the outcome of each hypothesis.

Table 4.3: Joint display: what each strand contributes to each proposition, and where they disagree

Proposition	Experiment	Interviews	Documents
A hidden fault raises deference	Deference 0.192 to 0.368; verification intent falls; confidence rises	Half describe tools that display no staleness signal at all	7 of 12 cases had no visible sign; those faults survived for years
Auditable framing raises deference with the data held constant	0.231 to 0.329, with no change in verification intent	96 per cent agree a report-backed decision is easier to defend; blame gap 2.23 points	Objection raised and overruled in 9 of 12 cases
The two failures are distinct	Additive, residual 0.015; different channels	Recording and answering distinguished in 4 transcripts	Detection controls present but authority controls absent
Deference narrows the option set	Not supported: IRR 0.94 [0.80, 1.10]	69 per cent agree; described at team level, not individual	Not addressable from documents
Answerability suppresses deference	Formal-route tick-box predicts nothing	Group describing a forum obliged to answer defers at 0.125 against 0.331	The one case where the objection carried, Zillow, was a voluntary commercial withdrawal, not a statutory route, which weakens rather than supports the proposition

Table 4.4: Outcome of each hypothesis

ID	Prediction	Estimate	Outcome
H1	Deference higher when the fault is hidden	OR 2.69 [1.77, 4.08]	Supported
H2	Deference higher under auditable framing, data constant	OR 1.81 [1.21, 2.70]	Supported
H3a	Higher stakes increase deference	OR 0.83 [0.71, 0.98] per point	Not supported
H3b	Higher stakes decrease deference	As H3a	Consistent, but stakes measured not manipulated
H4	Deferring managers name fewer alternatives	IRR 0.94 [0.80, 1.10]	Not supported
H5	Experience moderates H1	OR 1.03 [0.83, 1.28]	Null, but underpowered and exploratory

The strands agree that deference is conditional rather than general, and they agree on both conditions. They disagree about option narrowing, which is recorded as unresolved. The documentary strand supplies what neither of the other two can: evidence that these mechanisms produce serious harm at scale rather than small effects in a short online task.

Chapter 5

Discussion

5.1 What the evidence supports

Three propositions from Chapter 2 bear on different strands, and each is now set against the literature it came from.

Faults are behaviourally silent. Eleven of the twelve documented cases were coded silent or only partly visible, and silence goes with long survival. In the experiment, silence lowers intention to verify, raises stated confidence and nearly doubles deference, and thirteen of the 26 interview transcripts describe tools that display nothing when a feed fails. Data quality is therefore a problem of perception, not only of measurement. The dimensional tradition (Strong, Lee and Wang, 1997; Wang and Strong, 1996) describes believability as a dimension of quality; this study shows what believability does. A screen that carries no warning is read as an assurance, and the reduced sampling of raw evidence that Parasuraman and Manzey (2010) describe in operators appears in managers too. The result also extends Lyell and Coiera (2017): a hidden fault makes verification look unnecessary rather than merely difficult.

Legitimacy asymmetry is measurable. The construct comes from work on reactivity and performance management (Espeland and Sauder, 2007; Franco-Santos and Otley, 2018). Its signature appears in all five documented cases coded as strongly asymmetric, most sharply in Horizon, where system data supported criminal prosecutions against the testimony of those concerned (Post Office Horizon IT Inquiry, 2025). The interview material puts the construct on a scale. The gap between expecting to be held responsible for overruling the screen and for following it runs to more than two points on a seven-point measure, and it is positive for four participants in five. The accounts of professionals disengaging their own judgement where challenge is costly (Lebovitz, Lifshitz-Assaf and Levina, 2022) and of algorithms serving as instruments of control (Kellogg, Valentine and Christin, 2020) both describe the same asymmetry from other settings.

The accountability regime changes decisions without changing information. Only an experiment can test this, because it requires holding the data constant while varying how the decision will be reviewed, and the effect is present with the data unchanged. This is the prediction that legitimacy asymmetry makes and an attentional account does not; section 5.2 sets out what follows.

Two further results bear on sections 2.3 and 2.6. Higher perceived stakes went with slightly less deference, the direction that Dietvorst, Simmons and Massey (2015) would predict for a decision maker who checks more carefully when more is riding on the outcome, although the narrow range of stakes in the scenarios limits the claim. The experiment gave no support to the individual-level narrowing that Amabile (1998) and Doshi and Hauser (2024) would lead one to expect, while the interviews located the effect at the level of the team, where Woodman, Sawyer and Griffin (1993) place organisational creativity.

5.2 Revising the conceptual framework

Figure 2.2 assumed that reduced attention, the mechanism that Parasuraman and Manzey (2010) describe, is the main route by which dashboards displace judgement. That is at best half the story. In nine of the twelve documented cases somebody noticed the fault and said so, and what failed was turning that noticing into a decision.

The framework therefore splits into two failure modes, which Figure 5.1 adds.

Detection failure occurs when the fault is silent and nothing prompts the manager to think harder. Its causes are dashboard design and the visibility of the data quality signal, and its remedy is the disclosure of provenance and freshness on the screen.

Authority failure occurs when the fault is noticed and stated, but the objection has too little institutional standing to change the decision. Its cause is the accountability regime, and its remedy is not better information but a route by which an objection becomes something the process must record and answer. This mode is largely absent from the dashboard and automation bias literatures, which originate in single-operator settings where the person who notices is the person who decides.

The experiment shows that the split is a real distinction rather than a convenient one. The two manipulations do not merely produce separate effects; they move different measures. Hiding the fault lowers intention to verify and raises confidence, which is what an attentional mechanism looks like. Framing the decision as auditable moves neither, and still raises deference by about nine percentage points. A mechanism that leaves attention untouched and changes the choice is not a second route to inattention, and the framework has to carry both.

The two modes are also independent in the arithmetic: their effects add, with a residual close to zero. A dashboard can be fully transparent about data quality and still sit in a firm where disagreement carries career risk. Cognitive forcing functions (Buçinca, Malaya and Gajos, 2021) address detection and do nothing for authority.

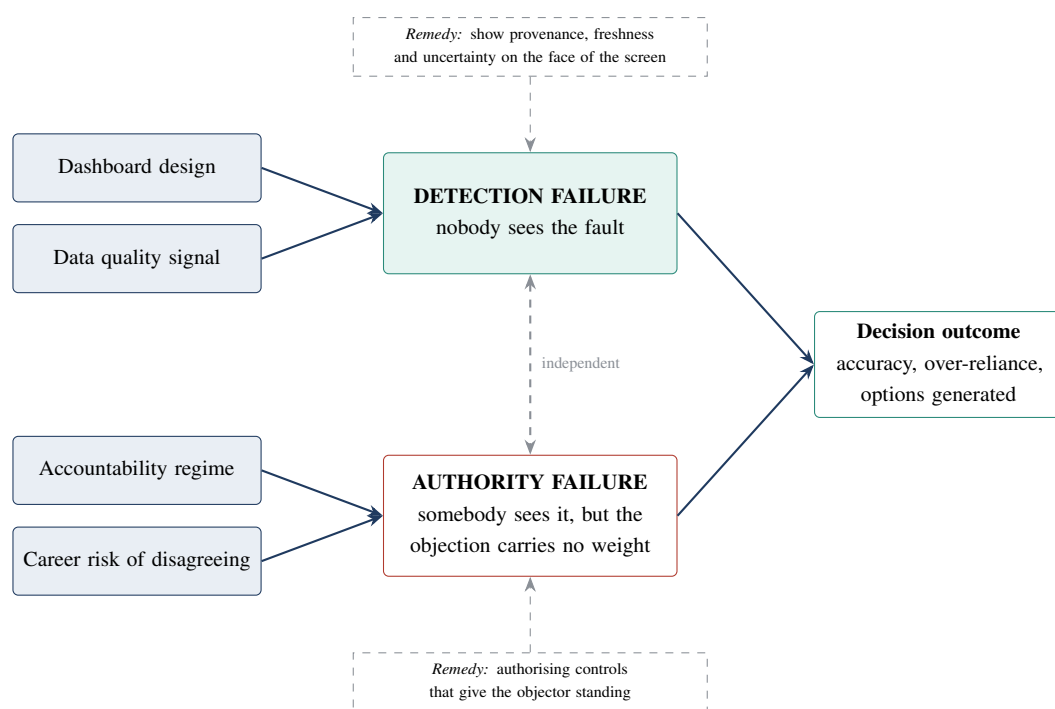


Figure 5.1: Revised framework separating detection failure from authority failure. The two have different causes and different remedies. Authority failure was the operating mode in nine of the twelve documented cases.

THINKING EXAMPLE 5 • Kovai Footwear

Two failures, not one

The revised framework separates cleanly on this case, which is the test of whether the distinction is doing any work.

Detection failure at Kovai is the dead counter. Its cause is that the screen reports a carried-forward reading identically to a live one. Its remedy costs one line of text, and Figure 1.2(b) shows what that line would have said.

Authority failure at Kovai is the capital expenditure form. Its cause is that the organisation decided, without ever deciding, which kinds of evidence have somewhere to be written down. Its remedy is not a better screen; fixing the banner alone would have left Meena in the same position.

The two are independent. Kovai happened to have both. An organisation can fix either one and still fail through the other, which is why the governance framework at section 6.2 has two limbs.

5.3 Where the strands disagree

An integration that reports only agreement has not been done properly. Two disagreements are on the record; the first is a failed prediction of this dissertation.

The prediction about relative size was wrong. The documentary strand was completed before the experimental data were analysed, and the reading of the documents led to a prediction: if authority rather than detection is the binding constraint, the accountability

manipulation should produce the larger effect. It did not. Hiding the fault returns an odds ratio of 2.69 and auditable framing 1.81, and in the decomposition detection accounts for about 62 per cent of the rise above baseline against 33 per cent for authority.

The reconciliation turns on how each strand selects its material. The documentary sample is drawn from failures serious enough to produce a statutory investigation. That selection favours organisations which had detective controls, since an organisation with no detection rarely generates the paper trail an inquiry needs, and it favours episodes in which an objection existed to be overruled. The experiment estimates the average effect of each condition across ordinary decisions, most of which never become failures. Both can be true at once: authority failure can dominate the cases that end in a public inquiry while detection failure is the larger effect across decisions in general. The claim is testable: authority failure should loom larger in high-consequence, heavily documented settings than in routine ones, and the interview participants who describe answerable objections do come disproportionately from regulated and safety-critical work.

The prediction was stated before the analysis and did not hold. The framework survives in a weaker form. Both modes operate, they are separable, and which one dominates depends on the setting rather than being a fixed property of dashboards.

The strands disagree about option narrowing. The experiment finds nothing: deference does not reduce the number of alternative actions named. The interviews describe narrowing directly, and most participants agree when the proposition is put to them. The most likely reconciliation is that the effect lives at the level of the team and the planning meeting, where the participants locate it, not in an individual's answer to a text box. It is recorded as unresolved, and section 6.6 treats it as a case for a different design rather than a larger sample.

5.4 What this evidence cannot establish

The documentary sample suffers from severe survivorship bias. Only failures that became public are observable, so a firm in which a manager questioned a dashboard and was listened to appears nowhere. Nothing at section 4.2 is evidence about how often managers defer; that strand establishes the anatomy of failure, not its rate. The cases are also heterogeneous, and the coding frame imposes a common structure on them, which is the usual trade-off in small-*n* comparative work between depth and comparability (Eisenhardt, 1989; Yin, 2018).

The experiment measures stated decisions where nobody loses money or standing, so the manipulation imitates accountability rather than imposing it. The bias runs towards understating the effect, which makes an observed effect the more notable and an absence uninformative.

The link between what managers say and how they behaved rests on 26 participants and is correlational, so a disposition to question a screen cannot be separated from a disposition to describe one's employer as an organisation that listens. Coding a construct such as legitimacy asymmetry also involves interpretation; the decision rules and the agreement between the free-text coding and an independent rating scale reduce the problem without removing it.

5.5 Implications

For theory. Legitimacy asymmetry is proposed as the mechanism that holds authority failure in place, and the interview material gives it a measurable form. The automation bias literature needs a second failure mode between noticing and deciding, because in organisations those two acts are often performed by different people.

For practice. Most organisational spending on analytics governance buys detective controls. On this evidence they address a real effect, and across ordinary decisions probably the larger one. No product reviewed at section 6.3 addresses authority failure, and organisations in the documentary record that had detective controls still failed through the other mode.

Chapter 6

Conclusion

6.1 Answers to the research questions

RQ1 asked in which situations managers defer to a dashboard, and what the deferring costs. The documentary record indicates that the worst outcomes go with a silent fault, an accountability regime that audits against the system, and a control architecture that detects without authorising. The experiment sizes the first two conditions. Deference runs from 0.147 of decisions where a warning is shown and the decision is the manager's own, to 0.421 where the fault is hidden and the decision is auditable, while the defensible action falls from 0.575 of cases to 0.271.

The part of RQ1 concerned with the range of options is answered negatively: deference does not reduce the number of alternative actions an individual manager names, although the interviews suggest that narrowing happens at team level.

RQ2 asked why managers defer, and why organisations keep it that way. The system output kept its priority not because contrary information was missing but because the objection carried less weight than the output it contradicted. In nine of the twelve documented cases somebody objected and was overruled. In the interviews, the gap between expected responsibility for overruling the screen and for following it runs to more than two points on a seven-point scale. In the experiment, one sentence describing how a decision will be reviewed raises deference by about nine percentage points and lowers defensible choice by about fourteen, while leaving intention to verify and confidence untouched, which is what defensibility predicts and attention does not.

Organisations keep it that way because the asymmetry is cheap to create and invisible once created. It requires no policy and no decision. It requires only the ordinary fact that a report is easier to point at afterwards than a conversation.

6.2 Towards a governance framework

The two failure modes produce a framework with two limbs. Table 6.1 sets out its form.

Table 6.1: Provisional governance framework for protecting human challenge

Failure mode	Type of control	What the evidence indicates in practice
Detection failure	Detective controls, which make the fault visible	Show data provenance and freshness on the face of the screen rather than in documentation. Reconcile record counts between source and destination, the absence of which explains the Public Health England undercount. Render a missing feed as a visible gap rather than as a lower number, as in Figure 1.2(b).
Authority failure	Authorising controls, which give standing to the person who disagrees	Provide a recorded and escalable route so that an objection becomes an auditable item alongside the dashboard reading. Require that overruling a recorded objection be itself recorded, with reasons. Separate the person who may raise a concern from the person whose performance the concern reflects on.
Both	Design and review	Treat any single-source input to a consequential automated decision as a control weakness. Ask whether reliance is being encouraged by framing decisions as auditable against the system.

THINKING EXAMPLE 6 • Kovai Footwear

What the two limbs would have cost

Limb one, detective. A nightly job comparing the count of counters expected against the count reporting, and a banner on the screen when they differ: perhaps a day of work for whoever maintains the extract. It would have shown Ramesh the line in Figure 1.2(b) on 13 May.

Limb two, authorising. One field on the capital expenditure form headed “local knowledge that conflicts with the Site Score”, and a rule that proceeding against anything written in it must be explained in writing by the person proceeding: perhaps an afternoon of work for whoever owns the form.

Neither is expensive. Neither existed. The second is the one nobody would have thought to build, because it does not look like an analytics problem.

The interview material supports a narrower claim than the table states. The participants who describe the asymmetry as absent or reversed do not describe better cultures. They describe specific arrangements: a personal signature that leaves no shelter in following the system, and a standing forum obliged to answer. What predicts behaviour is whether somebody must answer an objection, not whether it can be recorded, which is why limb two is written as a duty to reply rather than as a logging requirement.

One account cuts against the proposal and has to be answered. A technology director described an objection that was minuted and answered, attributed that outcome to their own seniority rather than to any mechanism, and observed that the formal route works well for the people who least need it. Seven transcripts carry that pattern, and those

participants deferred more than the rest. A route which relies on the objector's standing has not removed the asymmetry but relocated it. A control that meets the objection has to place the burden on the person proceeding: the requirement is not that an objection be well made, but that overruling one be explained in writing by whoever overrules it.

6.3 What current products do, and do not, address

An organisation reading section 6.2 will ask whether it can buy the answer. On the detective limb it largely can; on the authorising limb it cannot.

Detective controls are widely available commercially. Tableau attaches data quality warnings, including a stale-data type, and pushes them to a user opening a dependent view when an extract refresh fails (Tableau, 2026). ThoughtSpot verifies a Liveboard through a nominated reviewer (ThoughtSpot, 2026a) and surfaces dbt freshness metadata in Analyst Studio (ThoughtSpot, 2026b).

Two limitations follow. Most of this machinery attaches to assets rather than to the moment of decision, so a warning sits on a details tab and not where a manager under time pressure looks. And endorsement is not freshness: Microsoft Fabric offers certification badges, but an endorsement records a judgement made when the item was reviewed, not whether the numbers are correct today (Microsoft, 2025).

Authorising controls are not a product category. No tool reviewed here provides what limb two describes: a route by which an objection to a number becomes an auditable artefact beside that number, which somebody must answer in writing. Commenting exists in every product, but a comment is not a control, because it has no owner, no state and no requirement that anybody respond. Vendor investment runs the other way, towards agents that automate dashboard construction and semantic modelling (Avidon, 2025), which makes it faster to produce an authoritative-looking answer and no easier to doubt one. Appendix H sets the capabilities against the two failure modes. The gap is not carelessness. Detection failure is a property of a system, which a vendor can fix; authority failure is a property of an organisation's decision process, which no vendor can ship.

6.4 Contributions

Theoretical. The study separates detection failure from authority failure and proposes legitimacy asymmetry as the link between the accountability regime and the decision outcome, extending an automation bias literature that does not accommodate an organisational hierarchy standing between noticing and deciding.

Methodological. Selecting interviewees on observed experimental behaviour rather than on availability makes a sequential design explanatory in fact rather than only in name,

and grading documentary sources by evidential strength lets a reader discount the weaker cases.

Measurement. A negative result is offered as a contribution in its own right. A single item asking whether an organisation has a formal route for recording disagreement predicted nothing about behaviour, whereas measures asking whether an objection must be answered predicted it well. Survey work on analytics governance that asks the first question may be measuring the existence of policy documents.

Practical. Governance spending aimed only at better warnings addresses the mode that was not operating in most documented cases, and the other mode is not available to buy.

6.5 Limitations

The limits of the design are stated at section 3.8 and those of the evidence at section 5.4. The documentary coding was carried out by a single coder. Every judgement is anchored to a page-referenced quotation and can be re-checked, but the conventional check, a second coder on a subsample with agreement reported as Cohen's κ (Cohen, 1960) against the usual benchmarks (Landis and Koch, 1977), was not carried out, so the reliability of the coding is asserted rather than quantified. The instrument took most participants longer than the information sheet stated, which should be corrected in any replication. And the study examines a mechanism rather than a population, so it supports claims about how dashboard reliance works, not about how common it is.

6.6 Recommendations for further research

Four directions follow. A field study comparing decision episodes in one firm before and after a recorded objection route is introduced would test the authorising-control proposition directly. A study of option narrowing at the team level, observing what reaches a planning agenda rather than what an individual types into a box, would settle the disagreement at section 5.3. A replication in a generative artificial intelligence setting would show whether fluent system-generated text strengthens legitimacy asymmetry beyond a numeric dashboard. And a study sampling averted failures through near-miss registers would address the survivorship bias that no design based on public failure documents can escape. The detective versus authorising reading of failed controls at section 4.2 should also be coded as a dimension in its own right.

6.7 Concluding remark

The record of twelve organisational failures suggests that the risk has been described wrongly. The usual account says the danger is that managers fail to notice when a system is wrong. In most of these cases somebody did notice, and the failure was in what happened next. Ramesh had the same information in a two-line report from a colleague, and the form had nowhere to put it. The question is not whether dashboards should be used; plainly they should. It is how a manager remains a decision maker in an organisation where the screen has become the loudest voice in the room.

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Appendices

Appendix A. Literature Review Table

Table A.1 records the ten papers appraised in most detail, with the question each addresses, its method and finding, and the limitation that governs its use in this study.

Table A.1: Ten key papers, with critical appraisal and use in this study

Author (year)	Question addressed	Theory	Method	Key finding	Limitation and use here
Simon (1955)	How do people choose under cognitive limits?	Bounded rationality	Theoretical	Decision makers satisfice rather than optimise	Pre-dates digital decision aids; explains why an authoritative cue terminates search
Parasuraman and Manzey (2010)	Why do operators over-trust automation?	Attentional allocation	Integrative review	Reliance reduces attentional sampling of raw evidence	Aviation and process control settings with formal escalation; supplies the detection mechanism only
Dietvorst et al. (2015)	Do people abandon algorithms after error?	Algorithm aversion	Laboratory experiments	Observing error causes disproportionate abandonment	Student samples, generic forecasting, no accountability; supports H3b
Logg et al. (2019)	Do people prefer algorithmic or human advice?	Advice taking	Six experiments	General preference for algorithmic advice	Contradicts the aversion findings; motivates the conditional framing of RQ1
Burton et al. (2020)	When does algorithm aversion occur?	Augmented decision making	Systematic review	Reliance is condition dependent and not a stable trait	Synthesises heterogeneous tasks; core justification for the scenario design
Sarikaya et al. (2019)	What does dashboard design afford?	Visualisation design space	Design space analysis	Dashboard types differ sharply in what users may notice	Descriptive rather than behavioural; grounds the realism of the scenario dashboards
Wang and Strong (1996)	What does data quality mean to consumers?	Data quality dimensions	Two-stage survey	Quality is multi-dimensional and consumer defined	Built for assessment and not for predicting behaviour; defines the data quality manipulation
Nagle et al. (2020)	How well do firms measure data quality?	Managerial data quality practice	Practitioner study	Organisations systematically under-measure quality	Reports extent of under-measurement and not behavioural response; evidences the invisibility premise
Espeland and Sauder (2007)	How do public measures change organisations?	Reactivity	Qualitative field study	Measures reconstitute the reality they claim to describe	Single sector; theoretical basis for legitimacy asymmetry and for H2
Lebovitz et al. (2022)	How do professionals handle opaque AI?	Knowledge work and opacity	Comparative field study	Professionals disengage judgement where challenge is costly	Healthcare only; directly informs the interview protocol

Appendix B. Philosophical Positions Considered

The philosophical argument is stated at section 3.1. Table B.1 records the full comparison of the positions considered and the basis for the position adopted.

Table B.1: Philosophical positions considered, and the basis for the position adopted

Position	Implication for this study	Assessment
Positivism	Deference is an observable regularity to be measured and generalised.	Rejected. Captures the experimental strand but cannot accommodate legitimacy, which is neither directly observable nor stable across settings.
Interpretivism	Deference is meaning constructed by professionals in context.	Rejected as a sole position. Captures RQ2 but forbids the controlled comparison which RQ1 requires.
Critical realism	Observable deference is generated by underlying mechanisms operating under particular conditions.	Adopted. Licenses both measurement of the event and interpretation of the mechanism, and makes conditionality the object of study rather than a nuisance.
Pragmatism	Methods are warranted by their capacity to answer the question rather than by paradigm allegiance.	Adopted as the practical orientation, following established mixed methods guidance (Creswell and Plano Clark, 2018; Tashakkori and Teddlie, 2010).

Appendix C. Quality Criteria and Design Actions

The quality criteria are summarised at section 3.6. Table C.1 maps each criterion onto the specific action taken in the corresponding strand.

Table C.1: Quality criteria and the corresponding design actions

Strand	Criterion	Action taken
1	Internal validity	Randomised and counterbalanced vignette allocation; within-subjects design; manipulation check
1	Construct validity	Attention and comprehension checks; pilot testing of cue salience; explicit decision rule for the alternative-action count
1	External validity	Practising professionals rather than students; realistic dashboards; vignette stakes acknowledged as lower than real ones
1	Common method bias	Outcome is a behavioural choice rather than a self report; predictors are manipulated rather than measured (Podsakoff et al., 2003)
2	Documentary authenticity	Only independently verifiable primary documents admitted; evidential grading applied and reported
2	Coding reliability	Fixed extraction frame; verbatim page-referenced quotation for every judgement. Single coder; no inter-coder agreement statistic reported, see section 6.5
3	Credibility	Behaviour-linked sampling; member checking of summarised themes
3	Dependability	Audit trail of coding decisions and analytic memos; explicit decision rule for every code
3	Transferability	Thick description of participant sector, role and context
3	Reflexivity	The analytics training of the researcher creates a plausible bias towards reading deference as error; coding therefore actively seeks accounts in which deference proved correct

Appendix D. Design Alternatives Considered

The strategy is argued at section 3.3. Table D.1 records each alternative and the basis for rejecting it.

Table D.1: Design alternatives considered and why each was rejected

Alternative	Why it was rejected
Field experiment inside one firm	Strongest realism, but needs access to live decision systems, carries commercial confidentiality risk, and would mix the accountability manipulation with one firm's culture. Not feasible in the time available.
Cross-sectional survey of attitudes	Measures stated belief rather than decision behaviour, and is open to common method bias when predictor and outcome come from one instrument (Podsakoff et al., 2003).
Single organisational case study	Suited to identifying a mechanism (Eisenhardt, 1989; Yin, 2018) but cannot produce the conditional variation RQ1 needs, since conditions cannot be varied inside one setting.
Interviews alone	Answers RQ2 but not RQ1, and depends on managers recalling accurately a behaviour they have little reason to recall accurately.
Secondary analysis of existing data	No public dataset records dashboard-level decision episodes together with the contextual evidence available at the time.
Adopted: sequential explanatory mixed methods with a concurrent documentary strand	Vignettes vary the condition while holding the decision maker constant. Documentary analysis anchors the mechanism in real outcomes. Interviews explain the observed behaviour. Each strand covers a specific weakness in the others.

Appendix E. Experimental Factors, Measures and Power Estimation

Section 3.5.1 describes the instrument. Table E.1 records the factors and outcome measures as implemented, and Table E.2 the power grid on which the sample size rests.

Table E.1: Experimental factors, levels and how each is operationalised

Factor	Levels	How it is operationalised
Data quality signal (within)	Hidden or Visible	Presence or absence of a notice saying the feed is stale and incomplete, as in Figure 1.2
Accountability framing (within)	Auditable or Own judgement	Whether the decision will be recorded and reviewed against the dashboard, or recorded as the manager's own call
Scenario (within, crossed)	Eight, one per sector	Latin square: scenario j takes condition $(j + V) \bmod 4$, V drawn at random from 0 to 3
<i>Measured outcomes</i>		
Deference (primary)	Binary	Chose the dashboard-consistent action despite the conflicting local evidence
Decision accuracy	Binary	Chose the action defensible on all evidence available
Options generated	Count	Distinct alternative actions named; asked on a random three vignettes of eight
Confidence	Seven-point scale	Self-rated confidence in the action chosen
Perceived stakes	Seven-point scale	How much the participant judges to be riding on the decision
Verification intent	Binary	Whether more data were requested before deciding
Seconds on page	Continuous	Time from render to submission
Covariates: sector, years of experience, seniority, organisation size, dashboard use frequency, analytics familiarity, dissent route		

Monte Carlo power estimation. Generalised estimating equation, logit link, exchangeable working correlation clustered on participant. Participant standard deviation 0.85 on the log-odds scale (intraclass correlation about .18), scenario standard deviation 0.35, baseline $p = .40$, hidden effect +0.75 log-odds, auditable effect +0.95, interaction 0.50. Four hundred replications per cell; k is the number of vignettes per participant.

Table E.2: Estimated power (Monte Carlo, 400 replications per cell) by vignettes per participant and sample size

k	N	Observations	Power, main effect	Power, interaction	Note
4	60	240	.46	.13	
4	80	320	.60	.17	
4	100	400	.67	.18	
4	120	480	.74	.22	
6	80	480	.74	.19	same power, fewer recruits
6	100	600	.83	.28	
6	120	720	.89	.29	
8	60	480	.74	.22	same power again, half the recruits
8	80	640	.87	.29	design target
8	100	800	.94	.35	
8	120	960	.95	.40	interaction still underpowered

Two conclusions follow. Power tracks total observations and is close to indifferent to how they are split, so vignettes buy power more cheaply than recruits do. And the interaction never becomes well powered within any feasible design, which is why it is declared exploratory in advance. The achieved sample of 133 participants and 1,064 observations sits above the top row of the grid for the main effects.

Appendix F. Documentary Extraction Frame

The coding procedure is described at section 3.5.2. Table F.1 records the eight dimensions against which each of the twelve cases was coded.

Table F.1: Extraction frame used to code each documented failure

Dimension	What was coded
Failure type	Accuracy, timeliness, completeness, consistency or model logic
Signal visibility	Could the fault be seen from the screen or report at the point of use? Coded silent, partial or visible
Detection lag	Time between the fault starting and the organisation noticing
Human challenge	Was the output questioned? By whom? Was the objection escalated, ignored or overruled?
Legitimacy asymmetry	Evidence that system output was treated as more authoritative than a person's account; coded strong, moderate or weak
Control mechanism	What warning, override or reconciliation existed, and why it failed
Organisational impact	Financial, operational, regulatory and human consequences
Evidential grade	A or B, as defined at section 3.5.2

Appendix G. Disconfirmation Criteria for Each Proposition

The joint display is at Table 4.3 in section 4.5, where it sits with the discussion it supports. Table G.1 records what was fixed in advance as the evidence that would count against each proposition, and what the data showed.

Table G.1: What would have disconfirmed each proposition, and what was found

Proposition	What would disconfirm it	What was found
A hidden fault raises deference	A confidence interval for the hidden-fault odds ratio that includes 1, or intention to verify that does not fall when the notice is removed	OR 2.69 [1.77, 4.08]; verification intent fell. Not disconfirmed
Auditable framing raises deference with the data held constant	An accountability effect that disappears once scenario and perceived stakes are controlled, or one accompanied by a fall in intention to verify, which would make it a second attentional route rather than a distinct mechanism	OR 1.99 after adjustment; verification intent unchanged. Not disconfirmed
The two failures are distinct	A large positive interaction, indicating that the two conditions compound rather than add; or both manipulations moving intention to verify by similar amounts	Interaction OR 0.87, residual 0.015; only one manipulation moved verification. Not disconfirmed
Deference narrows the option set	Confirmation would require an incidence rate ratio below 1 with an interval excluding it	IRR 0.94 [0.80, 1.10]. Not supported at individual level
Answerability suppresses deference	Interview participants describing an answerable objection deferring at the same rate as the rest, or the rating measure R7 showing no association with behaviour	0.125 against 0.331; R7 correlates at $\rho = -0.66$. Not disconfirmed

Appendix H. Product Capability Against the Two Failure Modes

The argument is made at section 6.3. Table H.1 sets what current business intelligence and data observability products offer against the distinction between detection failure and authority failure.

Table H.1: Product capability set against the two failure modes

Capability	What is available	Which failure mode
Data quality warnings	Tableau: four warning types including stale data, high-visibility option, automatic warning on refresh failure	Detection
Verification badge	ThoughtSpot: verified Liveboard banner, plus a re-verification banner after any edit	Detection, partly
Freshness metadata	ThoughtSpot: dbt freshness surfaced in Analyst Studio reports	Detection
Endorsement	Microsoft Fabric: promoted, certified, master data. A judgement at review time, not a statement about today	Neither, despite appearances
Observability	Pipeline monitoring and anomaly detection upstream of the screen	Detection
Recorded dissent	Not found in any product reviewed. Comments exist everywhere; a comment has no owner, no state and no obligation to answer	Authority

Appendix I. The Twelve Documented Cases

The sample is described at section 4.2.1. Table I.1 lists each case with its sector, period, evidential grade and primary source.

Table I.1: The twelve cases, with evidential grade and primary source

ID	Case	Sector	Period	Gr.	Primary source
C1	Post Office Horizon	Postal and retail (UK)	1999–2019	A	Post Office Horizon IT Inquiry (2025), Vol. 1
C2	NATS flight planning	Aviation (UK)	2023	A	Civil Aviation Authority (2024), CAP2993
C3	Citibank risk data	Banking (US)	2013–2020	A	Office of the Comptroller of the Currency (2020), NR 2020-132
C4	PHE case undercount	Public health (UK)	2020	B	Official statements and parliamentary scrutiny (2020)
C5	Robodebt	Welfare (AU)	2015–2020	A	Royal Commission into the Robodebt Scheme (2023)
C6	TSB migration	Banking (UK)	2018	A	Financial Conduct Authority (2022), Final Notice
C7	Knight Capital	Trading (US)	2012	A	Securities and Exchange Commission (2013), Order 34-70694
C8	Ofqual grading	Education (UK)	2020	A	Office for Statistics Regulation (2021)
C9	Dutch childcare benefits	Welfare and tax (NL)	2013–2020	A	Autoriteit Persoonsgegevens (2021)
C10	Goldman Sachs reporting	Banking (UK)	2007–2017	A	Financial Conduct Authority (2019), Final Notice
C11	Zillow Offers	Property technology (US)	2021	B	Zillow Group (2021), Q3 shareholder letter
C12	Boeing 737 MAX (MCAS)	Aerospace (US)	2012–2019	A	US House Committee on Transportation and Infrastructure (2020)

Appendix J. Cross-Case Coded Matrix

The coding is described at section 4.2.5 and shown as a grid at Figure 4.5. Table J.1 gives it in full.

Table J.1: Cross-case coded matrix for the documentary strand

ID	Failure type	Signal	Lag	Objection	Asym.	Control
C1	Model logic	Silent	16 yr	Raised, overruled	Strong	Present, ineffective
C2	Model logic	Partial	~4 hr	Raised, slow escalation	Moderate	Present, ineffective
C3	Consistency	Silent	7 yr	Raised, overruled	Moderate	Present, ineffective
C4	Completeness	Silent	8 d	Not raised	Moderate	Absent
C5	Model logic	Silent	5 yr	Raised, overruled	Strong	Present, bypassed
C6	Consistency	Visible	1 d	Raised, overruled	Weak	Present, ineffective
C7	Model logic	Partial	<2 hr	Alerts issued, ignored	Weak	Absent
C8	Model logic	Partial	4 d	Raised, overruled	Strong	Present, ineffective
C9	Model logic	Silent	7 yr	Raised, overruled	Strong	Absent
C10	Accuracy	Silent	9 yr	Not raised	Weak	Present, ineffective
C11	Model logic	Partial	6 mo	Raised, acted upon	Moderate	Present, effective
C12	Model logic	Silent	6 yr	Raised, overruled	Strong	Present, bypassed

Appendix K. Reference and Search Audit Register

Every source cited in this dissertation has been checked against a primary record, and the check is documented in the register supplied alongside this document as `Sharma_250559280_Reference_Register.xlsx`. The register has four sheets.

Sheet 1, References. One row per source: the full Harvard entry as it appears in the reference list; the source type; the DOI or stable URL; the verification route, being either the Crossref DOI endpoint or the publisher's or issuing body's own page; the verification outcome; where a correction was made, what was wrong; the section of this dissertation in which the source is used; and a short statement of its relation to the argument.

Sheet 2, Search strategy. One row per search: the database, the exact Boolean string in the form entered, the fields searched, the date-range and language limits, the date run, the number of records returned, the number screened on title and abstract, and the number retained.

Sheet 3, Screening flow. The counts at each stage, from records identified through duplicates removed, title and abstract screening, full-text assessment and exclusions with reasons, to records included.

Sheet 4, Verification log. The raw outcome of the reference check, including the exact query issued for each source.

Outcome of the verification pass

All 85 reference-list entries were checked. 54 were confirmed against the Crossref DOI record and 26 against the publisher's or issuing body's own page, these being books, regulator notices, inquiry reports, vendor documentation and one news article, none of which Crossref indexes. Four entries carried errors and were corrected: a book attributed to the wrong imprint for its year, a vendor documentation page dated two years later than the page itself states, a documentation page whose title had been expanded with the name of the product section, and a company shareholder letter whose cited URL resolved to a different exhibit of the same filing. Six further entries were made more complete, chiefly by adding a place of publication, a report number or a direct rather than a site-root URL.

Appendix L. Written Interview Instrument

Section 3.5.3 describes the format. The instrument is delivered inside the same web page as the experiment and is supplied alongside this document. It has four parts.

Preamble and consent. Purpose; participation is voluntary; pseudonymisation; consent to the use of anonymised quotations; consent to storage; an instruction not to disclose confidential, client or commercially sensitive information; the right to skip any question or withdraw.

Twelve open questions, delivered in three pages of four.

1. The dashboards or reports you use in a typical week, and what your eye goes to first.
2. One recent decision that started from a dashboard: what the screen showed, what you noticed, what you checked before deciding.
3. How you know whether the numbers are current and complete, and what would make you doubt them. Whether your tools display any warning when a feed has failed.
4. An occasion when a dashboard told you something you did not believe, and what you did.
5. What happened after you raised it: who responded, whether the objection was written down anywhere, and the outcome. "It went nowhere" is a useful answer, not a failed one.
6. What kind of evidence carries weight when a decision is questioned later, and whether a report-backed decision is easier to defend than a judgement-backed one.
7. The two ways a decision can go wrong: following the dashboard and being wrong, against overruling it and being wrong. Whether the consequences are the same, as things actually work.
8. A time a colleague closer to the ground disagreed with a report: where the disagreement went, and whether anyone recorded it.
9. Whether working from a dashboard changes the range of options considered, and which options never reach the table.
10. Generated for each participant from their own record. The item restates one of the vignettes that participant saw, preferring one in which the fault was hidden and they

followed the screen, quotes their actual choice back to them, and asks what would drive that choice at their own work. The generated stem is stored in the dataset field Q10_prompt.

11. What would have to be true, in the tool, the process or the culture, for people to push back on a dashboard more often.
12. Anything important not asked.

Eight rating statements on a seven-point agreement scale, reproduced at Table 4.2. R4 and R5 are a matched pair, and their difference is stored as `blame_asymmetry_R5_minus_R4`.

Background block. Country, sector, job title, level, years of experience, dashboard use frequency, number of people responsible for, whether tools display a staleness warning, formal BI training, age band and highest qualification. Sector, seniority and years are carried across from the experiment rather than asked twice.

Appendix M. Documentary Source Index

The following publication indices were searched systematically for Strand 2: Office of the Comptroller of the Currency enforcement actions; Financial Conduct Authority final notices; Information Commissioner's Office enforcement register; Autoriteit Persoonsgegevens enforcement decisions; United Kingdom statutory public inquiry report libraries; Australian royal commission publications; National Audit Office value for money reports; House of Commons select committee reports; United States Securities and Exchange Commission administrative proceedings; Civil Aviation Authority CAP publications; and listed company annual report restatement disclosures.

Appendix N. Literature Search Documentation

The literature search was conducted in Scopus, Web of Science, Business Source Complete and the ACM Digital Library, with citation chaining forward and backward from the papers retained. The exact Boolean strings, the fields searched, the limits applied, the dates run and the numbers returned, screened and retained are recorded sheet by sheet in the register described at Appendix K, so that the search can be re-run rather than merely described.

The five strings around which the review is organised map onto the five literatures at Figure 2.3: business intelligence value and analytics capability; dashboard design and visualisation; reliance on algorithms and automation bias; data quality in context; and accountability, control and creativity. Documentary sources for Strand 2 were located separately, by searching the publication indices of the issuing bodies rather than the academic databases, as recorded at Appendix M.

Two rules were applied throughout. A source located through a citation chain was retained only if it could be found independently in one of the four databases. And press coverage was used to locate a case but never as the basis for a coded judgement, which is the admissibility rule at section 3.5.2.

Appendix O. Survey Instrument and Dataset

The scenario instrument is supplied alongside this document as a self-contained web page, `Sharma_250559280_Survey_Instrument.html`. It contains the participant information sheet, the consent items, the screening question, the eight scenarios covering the 2×2 conditions in a Latin square with their dashboard images, the per-vignette measures, two instructed-response attention checks, the manipulation and suspicion items, the written interview and the background block.

Responses were exported to a single long-format file with one row per vignette observation, pseudonymised at export and held on University encrypted storage. The dataset is supplied with this document together with the analysis scripts `analyse.py`, `code_interviews.py` and `figures.py`, so that every table and figure in Chapter 4 and Appendix P can be regenerated from the data. Every row carries a `response_id` and the version of the instrument the participant saw, so the Latin square allocation can be checked.

Appendix P. Full Model Estimates

All models are generalised estimating equations with robust standard errors, clustered on participant, with an exchangeable working correlation unless stated. $N = 133$ participants, 1,064 observations. Coefficients are on the log-odds scale.

Table P.1: Primary model: deference on data signal, accountability framing and their interaction

Term	<i>b</i>	SE	OR	95% CI	<i>p</i>
Intercept	−1.761	0.174	0.17	[0.12, 0.24]	< .001
Fault hidden	+0.988	0.213	2.69	[1.77, 4.08]	< .001
Auditable framing	+0.591	0.206	1.81	[1.21, 2.70]	.004
Hidden × auditable	−0.137	0.282	0.87	[0.50, 1.51]	.628

Table P.2: Secondary outcomes: defensible choice and intention to verify

Term	<i>b</i>	SE	OR	95% CI	<i>p</i>
<i>Defensible choice</i>					
Intercept	+0.303	0.126	1.35	[1.06, 1.73]	.016
Fault hidden	−0.794	0.166	0.45	[0.33, 0.63]	< .001
Auditable framing	−0.683	0.172	0.51	[0.36, 0.71]	< .001
Hidden × auditable	+0.183	0.247	1.20	[0.74, 1.95]	.458
<i>Intention to verify</i>					
Intercept	+0.991	0.141	2.69	[2.04, 3.55]	< .001
Fault hidden	−0.765	0.175	0.47	[0.33, 0.66]	< .001
Auditable framing	−0.019	0.200	0.98	[0.66, 1.45]	.924
Hidden × auditable	+0.019	0.258	1.02	[0.61, 1.69]	.942

Table P.3: Covariate-adjusted model, with scenario fixed effects (scenario terms omitted)

Term	<i>b</i>	SE	OR	95% CI	<i>p</i>
Intercept	−1.992	0.287	0.14	[0.08, 0.24]	< .001
Fault hidden	+1.048	0.208	2.85	[1.89, 4.29]	< .001
Auditable framing	+0.688	0.202	1.99	[1.34, 2.96]	< .001
Hidden × auditable	−0.158	0.279	0.85	[0.49, 1.47]	.570
Perceived stakes (centred)	−0.042	0.088	0.96	[0.81, 1.14]	.631
Experience (centred)	−0.102	0.113	0.90	[0.72, 1.13]	.366
Senior manager or above	−0.101	0.299	0.90	[0.50, 1.63]	.737
Uses dashboards daily	+0.066	0.207	1.07	[0.71, 1.60]	.749

Table P.4: Exploratory and robustness models

Term	<i>b</i>	SE	OR	95% CI	<i>p</i>
<i>H5, expertise moderation (declared exploratory)</i>					
Fault hidden	+0.918	0.130	2.50	[1.94, 3.23]	< .001
Experience (centred)	−0.148	0.117	0.86	[0.69, 1.08]	.203
Hidden × experience	+0.031	0.112	1.03	[0.83, 1.28]	.782
Auditable framing	+0.511	0.122	1.67	[1.31, 2.12]	< .001
<i>Perceived stakes (measured, not manipulated)</i>					
Fault hidden	+0.897	0.127	2.45	[1.91, 3.14]	< .001
Auditable framing	+0.534	0.124	1.71	[1.34, 2.18]	< .001
Perceived stakes (centred)	−0.181	0.083	0.83	[0.71, 0.98]	.029
<i>Excluding participants who inferred the design (n = 79)</i>					
Fault hidden	+1.304	0.266	3.68	[2.19, 6.21]	< .001
Auditable framing	+0.773	0.238	2.17	[1.36, 3.45]	.001
Hidden × auditable	−0.666	0.337	0.51	[0.27, 0.99]	.048

Option counts. Poisson model clustered on participant, 399 observations: deference IRR 0.94 [0.80, 1.10], hidden IRR 0.99 [0.86, 1.15], auditable IRR 0.91 [0.79, 1.06]. Pearson dispersion 0.95, so no negative binomial fallback was required.

Pre-committed fallback. Exact binomial on within-participant differences: 76 participants deferred more often under the hidden condition, 20 more often under the disclosed condition, 37 tied, $p < .001$.

Appendix Q. Ethical Approval, Risk Assessment, GDPR and Participant Documents

The following documents were submitted with the NBS8062 research proposal and approved before primary data collection began, as required by section 1.6 of the ISO8007 guidelines. They are reproduced in full on the pages that follow.

1. Business School ethical approval form for student projects (Sections 1 to 9)

2. Risk assessment, no travel
3. GDPR data management assessment
4. Interview participant consent form
5. Interview participant information sheet

Business School - Ethical approval form for student projects

SECTION 1: Applicant Details

Student Name and Number:	Kartikey Sharma (250559280)
Student Email Address:	k.sharma3@newcastle.ac.uk
Programme of Study:	MSc Business Analytics
Module Code and Name:	NBS8062 Research Methods
Supervisor Name:	Dr Xinyue Hao
Supervisor Email Address:	xinyue.hao@newcastle.ac.uk

SECTION 2: Project Details

Project Title:	When Dashboards Override Judgement: How Business Intelligence Dashboards Influence Managerial Decision-Making, Error and Creativity
Project Synopsis: <i>Please describe here what your planned research is about, why it is important and how you seek to conduct it.</i>	This dissertation project examines how Business Intelligence (BI) dashboards and organisational analytics influence managerial decision-making, particularly in relation to decision accuracy, over-reliance, data-quality risk and creative exploration. The research is important because dashboards increasingly act as decision infrastructure in operations, logistics, supply chain management, financial services, consulting and analytics environments. Managers may rely on visualised KPIs and analytical outputs when making decisions, even when contextual judgement or data-quality warnings suggest caution. The study will use a low-risk sequential explanatory mixed-methods design. Phase 1 will involve a web-based dashboard interpretation experiment/survey with approximately 50-80 adult professionals who use dashboards, KPI reports, BI tools or organisational analytics in their work. Participants will complete short hypothetical dashboard scenarios and decision questions. Phase 2 will analyse publicly available regulatory, organisational and post-incident evidence related to analytics or data-quality failures. Phase 3 will involve 6-8 online semi-structured interviews with consenting professionals to understand how and why dashboard reliance occurs. Participation will be voluntary, conducted online, and limited to adult professionals. No children, vulnerable groups, NHS patients, service users, confidential client records or sensitive organisational data will be involved. Interviewees will be reminded not to disclose proprietary, client-sensitive or employer-confidential information. Data will be anonymised/pseudonymised and stored securely on Newcastle University approved storage.

SECTION 3: Project Type

	YES	NO
--	-----	----

This research involves primary research.	X	
This research involves secondary research.	X	

*If you ticked **YES**, please complete the remainder of the form and append a copy of the information sheet and consent form for your research.*

*If you ticked **NO**, please sign the form on the last page and submit it to your supervisor for sign-off.*

	YES	NO
This research involves travel.		X

*If you ticked **YES**, please complete and append the risk assessment form 'travel'.*

*If you ticked **NO**, please complete and append the risk assessment form 'no travel'.*

SECTION 4: NHS, Health & Social Care: Facilities, Staff & Patients

	YES	NO
"Will the study involve participants recruited by virtue of being service users, their dependents, their carers' or human tissues or the use of NHS & Health / Social Care Facilities?"		X

*If you ticked **YES**, Please discuss alternative research questions / settings with your supervisor as a matter of urgency. The School does not support student projects involving patients of the NHS (or another health system in the UK or abroad) or their relatives.*

*If you ticked **NO**, please go to Section 5.*

SECTION 5: Human Participants in a Non-Clinical Setting

Does the research involve human participants, e.g. use of questionnaires, interviews, focus groups, observation, surveys or lab-based studies involving human participants? (If you are unsure please tick 'Yes' and complete the sub-questions.)	YES X	NO
---	-----------------	----

*If you ticked **YES**, please complete the sub-questions below and provide details.*

*If you ticked **NO**, please go to Section 6.*

Does the study involve any of the following?	YES	NO
a. The study involves children, or other vulnerable groups (inc. DBS should students want to work in schools), defined in Section 59 of the Safeguarding Vulnerable Adults Act 2006 as those who are relatively or absolutely incapable of protecting their own interests, or those in unequal relationships, e.g. participants who are subordinate to the researcher in a context outside the research ¹ ?		X

b. The study requires the co-operation of a gatekeeper for initial access to the groups or individuals to be recruited (e.g. organisational representative, chair / committee of a professional association)? Please note: gatekeepers are normally required when a case-study approach is taken.		X
c. It is necessary for participants to take part in the study without their knowledge and consent e.g. covert observation of people in non-public places?		X
d. Deliberately misleading participants in any way?		X
e. Psychological stress, anxiety, harm or negative consequences beyond that encountered in normal life?		X
f. Prolonged or repetitive testing i.e., more than 4 hours' commitment or attendance on more than two occasions?		X
g. Financial inducements (other than reasonable expenses and compensation for time)?		X

If you have answered **YES** to any of the sub-questions in Section 5, please provide details about the research, the issues that you have identified and the actions through which you seek to mitigate against them. Particular attention should be given to the procedures by which informed consent (the form should include the student's and supervisor's details to protect both researcher and interviewees) is sought and by which participants can withdraw from the research. **Please append a copy of the information sheet and consent form for your project.** The text box can be increased in size.

N/A - all Section 5 risk sub-questions are marked NO. The project involves adult professional participants only, recruited through LinkedIn, university networks and professional contacts. Participation will be voluntary and based on informed consent. The survey/experiment will include an electronic information and consent statement before participants proceed. Interview participants will receive an information sheet and consent form before the interview. Participants may withdraw at any time without penalty and may skip any question. The interview questions will focus on general professional experiences of BI dashboards and organisational analytics, not on confidential employers, clients, proprietary models or sensitive personal matters.

Please note that only low-risk projects are supported. Your supervisor may demand alterations to your research to ensure that it is low risk before being able to sign off this form.

Please continue with the Section 6.

SECTION 6: Data

	YES	NO
Does the research involve the viewing, usage or transfer of Sensitive personal data as defined as by the Terrorism Act or data governed by statute such as the Official Secrets Act , commercial contract or by convention e.g. client confidentiality? (If you are unsure please tick 'Yes' and complete the sub-questions.)		X

If you ticked **YES**, please complete the sub-questions below.

¹ Research involving adults (16+) who cannot consent, must have NHS review even if it is not health related.

If you ticked **NO**, please go to Section 7.

	YES	NO
a. Will the study involve the sharing of sensitive data outside the European Economic Area ?		X
b. Will the study involve the collection or analysis of sensitive data which will be identifiable within the project outputs and could potentially cause harm?		X
c. Will the study involve the collection or analysis of personal data without explicit consent?		X
d. Will the study involve the collection, viewing or dissemination of materials which could be considered 'extremist' and or 'terrorism related'?		X

If you have answered **YES** to any of the sub-questions in Section 6, there is a presumption that dissertation/project proposals will need individual ethical clearance. This would normally be incorporated in the existing project/dissertation proposal approval process. In addition, ensure that data management protocols are in place to conform with [GDPR \(2018\)](#) legislation. To comply with GDPR – ensure that all data is securely transferred via an encrypted USB when off-site, confidential, anonymous, the right to be forgotten and data will be destroyed post study. Similarly, if students are conducting lone interviews they should conform to the 'Lone Worker' protocol. The text box can be increased in size.

N/A - the project will not involve sensitive personal data as defined by the Terrorism Act, the Official Secrets Act, commercial contract restrictions or client confidentiality. Basic contact details may be used only for arranging interviews and will be stored separately from research data. Interview recordings will be stored securely, used only for transcription, and deleted after transcription. Transcripts and survey data will be anonymised/pseudonymised before analysis and reporting.

Please note that only low-risk projects are supported. Your supervisor may demand alterations to your research to ensure that it is low risk before being able to sign off this form.

Please continue with Section 7.

SECTION 7: International Projects

	YES	NO
Will the research be conducted outside of the European Economic Area (EEA) or will it involve international collaborators outside the EEA?		X

If you ticked **YES**, please complete the sub-questions below.

If you ticked **NO**, please go to Section 8.

	YES	NO
--	-----	----

a. Will there be an increased risk to researchers and participants where they are working remotely?		X
b. Will the research involve political sensitivities (e.g. offend specific public interests or the rights and reputations of others)?		X
c. Will the work undertaken overseas be governed by standards which are NOT equivalent to those in the UK?		X

If you have answered YES to any of the sub-questions in Section 7, please provide details about the research, the issues that you have identified and the actions through which you seek to mitigate against them. You should abide by the University's 'Lone Worker' protocols in informing family or friends of your whereabouts when collecting data. Particular attention should be given to differing expectations between the UK and the country in question and any risks associated. The text box can be increased in size.

N/A - the research will be conducted online from the UK and will not require travel outside the European Economic Area or involve international collaborators outside the EEA. Recruitment will be managed remotely through professional networks and no fieldwork travel will take place.

Please note that only low-risk projects are supported. Your supervisor may demand alterations to your research to ensure that it is low risk before being able to sign off this form.

Please continue with the Section 8.

SECTION 8: Declaration

By signing this declaration I, <i>[please insert your name here]</i> : * certify that the information contained in this application is accurate. * certify that the research will be undertaken in line with all appropriate local standards and regulations. * certify that I have discussed this form with my project supervisor and that s/he agrees with the answers given. * will submit a revised form should the project design change after this form has been signed off.	
Name of student:	Kartikey Sharma
Signature of student:	Kartikey Sharma
Date:	5 May 2026

SECTION 9: Supervisor sign-off and comments

If any of the sub-questions in Sections 5 to 7 has been answered with "yes", please provide a written justification below as to why the project as a whole is still low-risk. Special attention should be given to the conduct of the research, particularly when human participants are involved, when a gatekeeper is required and/or the research takes place internationally. Details about discussions with the student about key ethical issues and their mitigation should also be provided here.

Supervisor to review/sign. Suggested low-risk justification: This is a low-risk dissertation project involving adult professional participants only. Data collection will be online, voluntary and consent-based. The student will not recruit vulnerable groups, NHS/service-user participants or participants in subordinate relationships. No travel or gatekeeper access is required. The study will not collect sensitive personal data or confidential organisational/client information. Survey data and interview transcripts will be anonymised/pseudonymised, recordings will be deleted after transcription, and all files will be stored securely on University-approved storage.	
Name of supervisor:	Dr Xinyue Hao
Signature of supervisor:	[Supervisor to sign]
Date:	[Supervisor to complete]

Please attach any risk assessment forms and other applicable documents (such as information sheet and consent form) and submit them for sign-off as advertised in the module guide.

Newcastle University Business School Risk Assessment – No Travel etc.		
Module Code and Title	NBS8062 Research Methods	
Project Title	When Dashboards Override Judgement: How Business Intelligence Dashboards Influence Managerial Decision-Making, Error and Creativity	
Project synopsis (can be copied from Ethics form)	This dissertation project examines how Business Intelligence (BI) dashboards and organisational analytics influence managerial decision-making, particularly in relation to decision accuracy, over-reliance, data-quality risk and creative exploration. The research is important because dashboards increasingly act as decision infrastructure in operations, logistics, supply chain management, financial services, consulting and analytics environments. Managers may rely on visualised KPIs and analytical outputs when making decisions, even when contextual judgement or data-quality warnings suggest caution. The study will use a low-risk sequential explanatory mixed-methods design. Phase 1 will involve a web-based dashboard interpretation experiment/survey with approximately 50-80 adult professionals who use dashboards, KPI reports, BI tools or organisational analytics in their work. Participants will complete short hypothetical dashboard scenarios and decision questions. Phase 2 will analyse publicly available regulatory, organisational and post-incident evidence related to analytics or data-quality failures. Phase 3 will involve 6-8 online semi-structured interviews with consenting professionals to understand how and why dashboard reliance occurs. Participation will be voluntary, conducted online, and limited to adult professionals. No children, vulnerable groups, NHS patients, service users, confidential client records or sensitive organisational data will be involved. Interviewees will be reminded not to disclose proprietary, client-sensitive or employer-confidential information. Data will be anonymised/pseudonymised and stored securely on Newcastle University approved storage.	
Confirmation	I confirm that no travel outside of the University will be undertaken for the purposes of this research project.	
Student		
Kartikey Sharma	Kartikey Sharma	5 May 2026
Supervisor		
Dr Xinyue Hao	[Supervisor to sign]	[Supervisor to complete]

GDPR DATA MANAGEMENT ASSESSMENT

Student Name: Kartikey Sharma			
Supervisor: Dr Xinyue Hao			
Title of project: When Dashboards Override Judgement: How Business Intelligence Dashboards Influence Managerial Decision-Making, Error and Creativity			
Documenting:			
	Yes	No	
I understand that I must retain all my data until after I have graduated	X		
I will provide a methodology that explains how my data was collected	X		
Please explain how you will label and organise your data, records and files			
Files will be organised in a dissertation ethics folder on Newcastle University approved storage. Naming conventions will be: SurveyData_YYYYMMDD, DashboardExperiment_Responses_YYYYMMDD, Interview_P01_Audio_YYYYMMDD, Interview_P01_Transcript_Anon_YYYYMMDD, Consent_P01_YYYYMMDD and SecondaryCaseMatrix_YYYYMMDD. Participant identifiers will use codes such as P01, P02 etc. The key linking participant names/contact details to codes will be stored separately from the research data. Data will be retained until after graduation/assessment completion and then securely deleted in line with University guidance.			
Storing:			
	Yes	No	
I understand that I must keep all data and documentation related to my dissertation on the University server	X		
I will protect personal or sensitive data	X		
Please explain how you will protect personal or sensitive data (if you are not collecting this sort of data, please state this here)			
The study will avoid collecting sensitive personal data. Survey/experiment responses will be anonymous where possible. Interview contact details and consent forms will be stored separately from anonymised transcripts. Interview recordings will be password-protected, stored only on University-approved storage, used for transcription and deleted after transcription. Transcripts will remove names, employer names, client names, job titles where identifying, and any accidental references to confidential organisational information. Only the student, supervisor and authorised markers will access the data.			
Please explain how you will transfer and store data that is collected with mobile devices (e.g. audio recorder, mobile phone, USB stick, laptop)			
Data will primarily be collected online through survey software and online interviews. If an interview recording is temporarily stored on a laptop or recording device, it will be transferred to Newcastle University approved cloud storage within 24 hours and deleted from the local device immediately after successful transfer. No unencrypted USB storage will be used. Files will be password-protected where possible and accessed only from the student's password-protected device/account.			
Confidentiality, ethics and consent:			
	Yes	No	N/A
I have completed the ethics assessment	X		
I will follow this assessment throughout my dissertation and seek additional advice from my supervisor as needed	X		
I have gained consent from my participants, including their permission to share data with supervisor/second marker	X		
I have anonymised my data to remove identifying/personal information	X		
Copyright			
	Yes	No	N/A
Have you established who owns the copyright in your data?	X		
If you are re-using someone else's data sources, have you considered copyright?	X		
Please explain how you have accessed any secondary data being used and any associated copyright implications			
Primary data will be generated through the student's own survey/experiment and interview data collection. Secondary data will be limited to publicly available regulatory reports, annual reports, inquiry reports, enforcement releases and other publicly accessible documents. These sources will be used for academic analysis only, cited appropriately, and not reproduced beyond fair dealing/academic quotation limits. No proprietary databases or confidential company archives will be used unless explicit permission is obtained.			

Interview Participant Consent Form

When Dashboards Override Judgement: How Business Intelligence Dashboards Influence Managerial Decision-Making, Error and Creativity

Thank you for your interest in taking part in this research. Please complete this form after you have read the Information Sheet and/or listened to an explanation about the research study. You will be given a copy of this Consent Form for your records.

<i>Please tick box to confirm consent</i>		
1.	I confirm that I have read the information sheet for the above study, I have had the opportunity to consider the information, ask questions and I have had any questions answered satisfactorily.	
2.	I understand that my participation is entirely voluntary and that I am free to withdraw at any time without giving any reason. I understand that if I decide to withdraw, any data that I have provided up to that point will be omitted.	
3.	I consent to my anonymised research data being stored securely and used in follow-up publications.	
4.	I consent to being audio or video recorded and understand that the recordings will be stored securely on password-protected software and used for research purposes only. Recordings will be destroyed immediately after transcription.	
5.	I agree to take part in this research project.	
<i>Participant</i> _____ _____ _____ <i>Name of participant</i> <i>Signature</i> <i>Date</i>		
Researcher Signature Date Name of researcher Kartikey Sharma Kartikey Sharma 5 May 2026		

Interview Participant Information Sheet

Title: When Dashboards Override Judgement: How Business Intelligence Dashboards Influence Managerial Decision-Making, Error and Creativity

Invitation and Brief Summary

You are being invited to take part in a research study for an MSc Business Analytics dissertation at Newcastle University Business School. Before deciding whether to take part, please read this information carefully and ask any questions you may have. Taking part is entirely voluntary. If you decide to participate, you will be asked to sign a consent form before the interview. You may withdraw at any time without giving a reason and without penalty.

What is the purpose of the research?

The purpose of this study is to understand how Business Intelligence (BI) dashboards and organisational analytics influence managerial decision-making. The research explores how professionals interpret dashboard outputs, why they may rely on dashboards over contextual judgement, and whether dashboard design or data-quality issues contribute to decision errors or reduced creative exploration.

Why have I been invited?

You have been invited because you are an adult professional who uses, or has recently used, dashboards, KPI reports, BI tools or organisational analytics in a work setting such as operations, logistics, supply chain management, finance, consulting, technology or analytics.

What does taking part involve?

Participation involves one online semi-structured interview, normally conducted through Microsoft Teams, Zoom or another agreed online platform. The interview will last approximately 30-40 minutes. With your permission, the interview may be audio or video recorded for transcription. You can refuse recording and still discuss whether written notes can be taken instead.

What will the interview be like?

The interview will ask about your general professional experience of using dashboards and analytics in decision-making. Questions may cover what you notice first in a dashboard, how you decide whether to trust or challenge a dashboard, how data-quality warnings are treated, and whether dashboards encourage or constrain alternative ideas. You will not be asked to disclose confidential employer, client, financial, proprietary or commercially sensitive information. Please avoid naming clients, employers, colleagues or confidential systems in your answers.

What will happen to the information I provide?

Interview recordings will be stored securely on password-protected Newcastle University approved storage and used only for transcription. Recordings will be deleted after transcription. Transcripts will be anonymised/pseudonymised by removing names, employer/client identifiers, and other details that could reasonably identify you or your organisation. Anonymised extracts may be used in the dissertation and related academic outputs.

Can I withdraw?

Yes. You may withdraw before or during the interview without giving a reason. After the interview, you may request withdrawal of your data up to the point at which the data have been anonymised and incorporated into analysis. After that point, it may not be possible to identify and remove your individual data.

What are the possible benefits of taking part?

There are no direct personal benefits. However, your participation may help improve understanding of how organisations can use dashboards and analytics while preserving human judgement, data-quality challenge and creative decision-making.

What are the possible disadvantages and risks of taking part?

The study is low risk. Some questions may ask you to reflect on workplace decision-making, but you can skip any question or stop the interview at any time. You should not disclose confidential, commercially sensitive or personal information. Any accidental identifying information will be removed during transcription/anonymisation.

Confidentiality and anonymity

Your identity will be protected as far as reasonably possible. Interview transcripts will use participant codes such as P01 and P02 rather than names. Contact details and consent forms will be stored separately from research data. Only the student researcher, supervisor and authorised markers will have access to the research data.

Who is funding this research?

This research is not externally funded. It is being conducted as part of the MSc Business Analytics dissertation/research methods assessment at Newcastle University Business School.

Has this study received ethical approval?

This study will be submitted for ethical approval through Newcastle University Business School. No primary data collection will begin until ethical approval has been granted.

Who should I contact for further information?

Student researcher: Kartikey Sharma, Student ID: 250559280, [Email: K.sharma3@newcastle.ac.uk](mailto:K.sharma3@newcastle.ac.uk)

Supervisor: Dr Xinyue Hao, Email: xinyue.hao@newcastle.ac.uk

Participant Information Sheet Version: 1.0, Date: 5 May 2026